

Cambridge IGCSE Economics 0455

经济学全册 · 双语深度学习教材

2026 syllabus edition · 6 units · 33 lessons

本书吸收参考理科全册讲义“单元推进、概念-例子-题解相邻、版式适合打印”的优点，并按经济学的分析方法重新设计：每课从定义和机制出发，以数据、图表、因果链和原创 Cambridge-style 题目完成迁移。

依据 Cambridge International Economics 0455 syllabus for examination in 2026 及公开样卷/评分材料编写。题目、数字、情境和图表均为原创，不复制受版权保护的历年试题。2027 起 syllabus 与考试时长、题量有调整，使用者应按实际报考年份核对。

全册结构与使用方法 | How this book is organised

全册按官方 6 个主题组织为 33 课。知识点数量由每课实际概念与计算负担决定，不套固定模板；正文例题用于演示方法，逐题精讲使用新情境检验迁移，并把答案直接放在题目后。英文不是单独附加栏，而是嵌入定义、机制、图解和 model answer。

图表学习顺序：先写坐标轴与单位，再标曲线和原状态，接着画移动箭头和新状态，最后用完整英文说明 cause → curve movement → disequilibrium/adjustment → final result。没有图表要求的课不强行配图。

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Unit 1 | 基本经济问题

The basic economic problem

单元导读 | Unit introduction

从有限资源出发，依次理解 scarcity、choice、opportunity cost、factors of production、PPC 与 economic systems。本单元不是背四个定义，而是建立后续所有分析共用的逻辑：资源有限，所以必须选择；选择会放弃最佳替代方案；生产能力与资源配置可以用 PPC 表示。

学习主线 | Learning route

先学会把生活情境翻译成经济变量，再判断题目是在改变实际产量、资源配置还是最大生产能力。PPC 的点、移动和曲线位移必须分开；市场、混合和计划经济的比较则围绕 information、incentive、efficiency 与 equity。

English overview

Start with the constraint. Scarcity forces households, firms and governments to choose. Every feasible choice has an opportunity cost. A PPC then shows the maximum combinations that can be produced with given resources and technology. The unit finishes by comparing how different economic systems answer what, how and for whom to produce.

本单元课时 | Lessons in this unit

第 1 课 | 2.1 稀缺、选择与机会成本 (Scarcity, choice and opportunity cost)；第 2 课 | 2.2 四种生产要素与报酬 (Factors of production)；第 3 课 | 2.3 生产可能性曲线 (Production possibility curve, PPC)；第 4 课 | 2.4 经济制度与资源配置 (Economic systems)；第 5 课 | 2.5 逐步判断：一个 PPC 题到底在问什么？

第 1 课 | 2.1 稀缺、选择与机会成本 (Scarcity, choice and opportunity cost)

所属大章: 基本经济问题 | The basic economic problem

知识点精讲 | Core knowledge

稀缺 (scarcity) 不是 “东西很少” , 而是资源不足以同时满足所有欲望。空气通常不稀缺, 但在潜水瓶、医院氧气或严重污染情境中会变成稀缺资源。稀缺迫使家庭、企业、政府作出选择 (choice) 。每一次选择的真正代价是机会成本 (opportunity cost) : 被放弃的 “最佳替代方案” 的价值, 而不是所有放弃物的总和。

例子: 周六下午你可做兼职赚 120 元, 或复习经济学。若选择复习, 机会成本是 120 元, 而非 “120 元加上休息时间” 。答题时写: The opportunity cost of studying is the income forgone from the best alternative use of time.

深度展开与真题化应用 | In-depth development (共 6 个知识点)

知识点 01/06 | 稀缺、选择与机会成本: 概念边界与生活化理解 | Concept boundaries in everyday choices

稀缺必须同时包含两个条件: 人的欲望很多, 而可用于满足欲望的资源有限。因此, 价格很高不等于稀缺, 免费物品也不一定永远不稀缺。学校只有一间实验室, 政府只有一笔预算, 家庭一天只有 24 小时, 三者都面对资源约束。选择发生后才出现机会成本, 而且机会成本只计算 next best alternative forgone (被放弃的最佳替代方案) 。

知识点 02/06 | 稀缺、选择与机会成本: 从选择推到经济后果 | From a choice to its economic consequence

假设市政府可用 2

亿元修建医院或体育馆。若医院是最佳替代用途, 选择体育馆的机会成本就是医院带来的医疗服务, 而不是 “2 亿元” 本身, 因为这笔钱在两种方案中都会使用。选择还可能改变不同群体的福利: 体育馆使用者受益, 等待医疗的人可能受损。IGCSE 题目中的高分答案不仅写定义, 还要把有限资源、互斥用途和最佳替代方案连接起来。

知识点 03/06 | 稀缺、选择与机会成本中的主体、变量与判断对象 | Agents, variables and object of analysis

家庭面对 income 与 time, 企业面对 labour、capital 和 finance, 政府面对 tax revenue 与公共资源。三者都必须从互不相容的用途之间选择; 机会成本必须从该主体实际可行的替代方案中确定。

知识点 04/06 | 稀缺、选择与机会成本的因果机制 | How the mechanism works

政府税收有限 -> 同一批建筑工人和材料不能同时建设医院与机场 -> 选择机场 -> 医院服务成为 opportunity cost -> 不同居民群体受到不同影响。

知识点 05/06 | 稀缺、选择与机会成本的关键英文表述 | Exam-ready explanation in English

Scarcity creates a constraint, choice is the decision made under that constraint, and opportunity cost measures the value of the next best option forgone. The alternative must be feasible and mutually exclusive with the chosen option. A complete answer identifies both the decision-maker and the specific benefit sacrificed.

知识点 06/06 | 稀缺、选择与机会成本最容易混淆的口径 | Common misconception

不要把 shortage 与 scarcity 混用。Shortage 是某价格下 $Q_d > Q_s$, 价格变化可消除; scarcity 是资源相对于欲望有限, 是所有经济体长期面对的基本问题。

逐题精讲与评价拓展 | Exam workshop (共 2 项)

精讲 01/02 | 选择题: 区分最接近的概念 | Multiple-choice application

Question. A council can build a library, repair a road or leave the money in a bank. It builds the library; road repair was its second preference. What is the opportunity cost?

正确答案 | Correct answer. The benefit from repairing the road. The bank deposit is another alternative but is not the next best one.

精讲 02/02 | 结构题: 把情境发展成完整分析 | Structured application

Question. Explain why economic growth does not remove scarcity.

参考答案 | Model answer. Growth shifts productive capacity outwards and allows more wants to be satisfied. However, human wants can continue to expand and resources remain limited relative to those wants. The economy therefore still has to choose how to use labour, land and capital, so opportunity cost remains.

第 2 课 | 2.2 四种生产要素与报酬 (Factors of production)

所属大章: 基本经济问题 | The basic economic problem

知识点精讲 | Core knowledge

Land (土地): 自然资源, 如耕地、矿产、森林; 报酬为 rent (地租)。

Labour (劳动): 人提供的体力与脑力; 报酬为 wages (工资)。

Capital (资本): 用于生产的人造物品, 如机器、厂房、软件; 报酬为 interest (利息)。它不是“有钱”本身。

Enterprise (企业家才能): 组织资源、承担风险、创新和作决策; 报酬为 profit (利润)。

常见误区: CEO 的工作属于 labour; 一辆消费者自用汽车不是 capital, 但出租车公司用于提供运输服务的汽车是 capital。

深度展开与真题化应用 | In-depth development (共 7 个知识点)

知识点 01/07 | 四种生产要素与报酬: 四种要素不是四种“东西” | Classifying factors by their productive function

Land 包括自然提供的资源, 不只是一块土地; 海洋、太阳能、矿藏都属于 land。Labour 是人的体力和脑力投入, 数量之外还要看技能、健康和教育。Capital 是为生产其他产品而制造的生产资料, 银行账户里的钱是 finance, 不自动等于 capital。Enterprise 负责组织其他要素、承担不确定性并作出创新决策。同一个物品的分类取决于用途: 家庭自用电脑是消费品, 设计公司用于工作的电脑是 capital good。

知识点 02/07 | 四种生产要素与报酬: 数量、质量、流动性与报酬 | Quantity, quality, mobility and factor rewards

生产能力不仅由要素数量决定, 还取决于质量和 mobility。培训提升 labour quality; 新机器提高 capital quality; 灌溉可提高 land productivity。工人可能因住房、资格或语言限制而不能从衰退行业转到扩张行业, 这叫 occupational/geographical immobility。要素报酬通常写作 land-rent、labour-wages、capital-interest、enterprise-profit, 但现实中企业家也可能领取工资, 考试应按其所承担的经济功能分类。

知识点 03/07 | 四种生产要素与报酬中的主体、变量与判断对象 | Agents, variables and object of analysis

Land 看自然资源, labour 看人的体力与脑力, capital 看人为制造且用于生产的工具, enterprise 看组织与承担风险。分类取决于资源在生产中的功能, 不取决于物体名称。

知识点 04/07 | 四种生产要素与报酬的数字或现实情境 | Numerical or real-world application

五名工人每天生产 200 件, labour productivity=200/5=40 件/人。培训后四名工人生产 200 件, 生产率=50 件/人, 提高 25%。总产量没变, 但单位劳动投入效率提高。

知识点 05/07 | 四种生产要素与报酬的关键英文表述 | Exam-ready explanation in English

The classification of a resource depends on how it is used. A van owned by a delivery firm is capital, while the driver provides labour. Productivity is output per unit of input. An increase in the number of workers can raise total output without raising labour productivity.

知识点 06/07 | 四种生产要素与报酬最容易混淆的口径 | Common misconception

Money used to buy a machine is finance; the machine itself is capital. Shares and banknotes are financial assets, not automatically factors of production.

知识点 07/07 | 四种生产要素与报酬的综合应用 | Integrated application

增加工人数提高 labour quantity; 培训提高 labour quality; 同样的工人生产更多才是 productivity 提高。答题时分别说明数量、质量和每单位投入产出, 不能用“劳动力增加”同时代替三者。

逐题精讲与评价拓展 | Exam workshop (共 2 项)

精讲 01/02 | 选择题: 区分最接近的概念 | Multiple-choice application

Question. Which change definitely improves the quality of labour: more mineral deposits, worker training, a larger factory, or a lower interest rate?

正确答案 | Correct answer. Worker training, because it increases human capital and the skills embodied in labour.

精讲 02/02 | 结构题: 把情境发展成完整分析 | Structured application

Question. Analyse how an increase in the quantity and quality of capital may affect an economy.

参考答案 | Model answer. More and better machinery can raise output per worker. Higher productivity may lower average cost, enabling firms to reduce prices or increase profit. Competitiveness and exports may rise, increasing real GDP. The effect depends on workers having the skills to operate the capital and on sufficient demand.

第 3 课 | 2.3 生产可能性曲线 (Production possibility curve, PPC)

所属篇章：基本经济问题 | The basic economic problem

知识点精讲 | Core knowledge

PPC 显示在既定资源与技术下，一个经济体能够生产两种产品的最大组合。曲线上任一点是 productively efficient (生产有效率)；曲线内表示资源闲置或失业；曲线外短期不可达到。曲线向外移动来自更多/更好的资源、教育培训、技术进步和投资；战争、自然灾害或劳动力流失会使曲线向内移动。

若 PPC 向原点凹，表示生产一种产品越多，转移资源的适配度越差，opportunity cost increases (机会成本递增)。画图题务必写坐标、两种产品、曲线与移动箭头；不要把“沿 PPC 移动”(产品组合改变)误画成“PPC 整条移动”(生产能力改变)。

PPC 图像的读法：从坐标到结论

先画两条互相垂直的轴：横轴写 Good X (例如 computers)，纵轴写 Good Y (例如 food)，并标出原点 O。曲线连接两个截距：若所有资源生产食物，位于纵轴截距；若所有资源生产电脑，位于横轴截距。曲线上的 A、B、C 是不同的可行组合，不是三个“效率等级”。从 A 沿曲线向 B 移动，表示资源从食物部门转向电脑部门；生产能力没有改变，只是组合改变。

图中曲线以内的点 D 代表 productive inefficiency (生产无效率)：在不减少另一种产品的情况下仍可增加产出，原因可能是失业、机器闲置或管理低效。曲线以外的点 E 在当前资源和技术下不可达。若培训提高劳动生产率，整条 PPC 向外移；若地震摧毁厂房，整条 PPC 向内移。答题时用箭头标示 shift，并明确写出“productive potential changes”。

为什么曲线通常向原点凹？机会成本的推演

假设经济体只有土地、劳动和资本，生产电脑和食物。最初从食物转出少量资源时，可以先转移最适合电脑生产的工人和设备，所以电脑增加 1 单位只牺牲 1 单位食物。继续转移时，电脑部门需要使用原本更适合种植的土地和农业工人，牺牲的食物增加到 2、3、4 单位。这就是 increasing opportunity cost (机会成本递增)，图上表现为越接近横轴，曲线越陡。

由表可见，每增加相同的 10 台电脑，放弃的食物越来越多。若题目要求“calculate opportunity cost”，要写 $\text{opportunity cost} = \text{output forgone} / \text{output gained}$ ，并说明是哪个区间；不能只写“机会成本为 14”，因为不同区间的数值不同。

课堂例题：区分 movement 与 shift

某国从组合 A (20 computers, 88 tonnes of food) 变为组合 B (30 computers, 74 tonnes of food)。第一问：发生了什么？答案是沿 PPC 移动，电脑增加 10，食物减少 14；机会成本为 14 吨食物。第二问：若新工厂和自动化使每一种产品的最大产量都提高，如何作图？答案是 PPC outward shift；原曲线上的 A、B 仍是旧生产能力下的组合，新曲线应位于右上方。

高分英语表达：The economy moves from A to B along the same PPC, so resources are reallocated and the opportunity cost of 10 additional computers is 14 tonnes of food. In contrast, an outward shift would indicate an increase in productive potential caused by better resources or technology.

画图与评价的评分清单

1. 坐标轴有名称和方向；2. 曲线标为 PPC/PPC；3. 点标 A、B、inside、outside；4. 移动用沿曲线箭头，能力变化用整条曲线箭头；5. 解释原因，而非只写“向外”；6. 写出短期可行性和长期增长的区别。若资源只适合其中一种产品，PPC 可接近直线；若资源完全不能互换，机会成本近似不变。

深度展开与真题化应用 | In-depth development (共 9 个知识点)

知识点 01/09 | 生产可能性曲线: PPC 是约束图, 不是需求供给图 | The PPC as a production constraint

PPC 的每一个点都对应两种产品的产量组合, 并假定资源数量、资源质量和技术暂时不变。曲线上点表示所有现有资源已被充分有效利用; 曲线内点表示实际产量低于潜在产量, 原因可能是 cyclical unemployment、机器闲置或组织低效; 曲线外点不是“永远不可能”, 而是以当前生产能力不能持续达到。贸易可以让消费点位于本国 PPC 外, 但并不等于本国生产点在曲线外。

知识点 02/09 | 生产可能性曲线: 机会成本、曲率与偏向性移动 | Opportunity cost, curvature and biased shifts

从 A(0 computers, 100 food) 到 B(10, 96), 10 台电脑的机会成本是 4 单位 food; 再到 C(20, 88), 新增 10 台电脑的机会成本变为 8。机会成本递增的原因不是曲线“天生弯”, 而是生产要素具有 specialised uses。若只有电脑技术进步, PPC 主要绕食物轴截距向外旋转; 若教育普遍提高生产率, 两端都可能外移。沿曲线移动表示 resource reallocation, 不表示 growth。

知识点 03/09 | 生产可能性曲线中的主体、变量与判断对象 | Agents, variables and object of analysis

坐标轴是两种产品, 边界由现有资源、技术和生产率决定。点的位置回答 actual output, 曲线位置回答 productive potential, 曲线斜率所表示的取舍回答 opportunity cost。

知识点 04/09 | 生产可能性曲线的因果机制 | How the mechanism works

更多资本品生产 → 当前消费品机会成本上升 → capital stock 随时间增加 → future productivity rises → PPC later shifts outwards → future consumption possibilities expand。

知识点 05/09 | 生产可能性曲线的数字或现实情境 | Numerical or real-world application

组合 A 为 0 台机器和 100 吨粮食, B 为 10 和 94, C 为 20 和 84。A 到 B 的 10 台机器成本 6 吨粮, B 到 C 成本 10 吨, 显示 increasing opportunity cost。

知识点 06/09 | 生产可能性曲线的图表与数据读法 | Diagram and data interpretation

先读两条坐标轴代表的产品, 再判断点在曲线上、曲线内还是曲线外。沿同一条 PPC 移动表示产品组合改变; 资源数量、资源质量或技术改变才使整条 PPC 移动。图形结论必须同时写出 gained output 与 forgone output。

知识点 07/09 | 生产可能性曲线的关键英文表述 | Exam-ready explanation in English

A bowed-out PPC reflects specialised resources. As production of one good expands, resources increasingly unsuitable for that activity are transferred, so more of the other good must be sacrificed. A straight line indicates constant opportunity cost.

知识点 08/09 | 生产可能性曲线最容易混淆的口径 | Common misconception

曲线上的所有点都是 productively efficient, 但只有符合社会偏好的组合才是 allocatively efficient。两者不能互换。

知识点 09/09 | 生产可能性曲线结论成立的条件 | When the conclusion may change

PPC 外移不保证生活水平同比提高。人口增长、污染、分配不均以及新增能力是否实际使用都会改变最终结果。

逐题精讲与评价拓展 | Exam workshop (共 3 项)

精讲 01/03 | 选择题: 区分最接近的概念 | Multiple-choice application

Question. An economy moves from a point inside its PPC to a point on the curve. What is the most likely explanation?

正确答案 | Correct answer. Unemployment or inefficiency has fallen. Existing productive capacity is used more fully; the PPC itself need not shift.

精讲 02/03 | 结构题: 把情境发展成完整分析 | Structured application

Question. Explain how investment in capital goods can affect current and future living standards.

参考答案 | Model answer. Allocating more resources to capital goods requires less current production of consumer goods, so present consumption may fall. The larger capital stock can raise productivity and productive potential later, shifting the PPC outwards. Future real income and consumption may then increase, provided the capital is useful and fully employed.

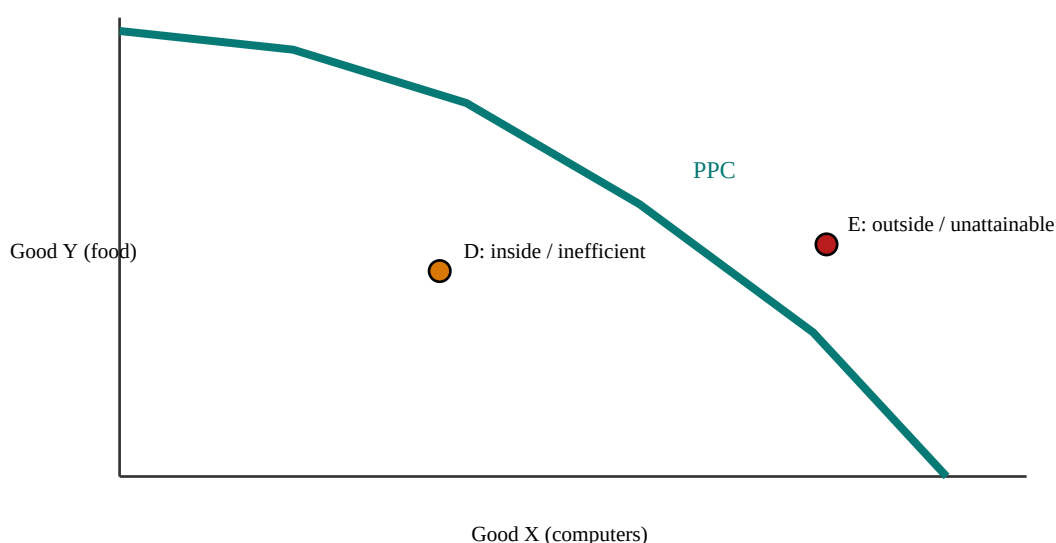
精讲 03/03 | 图表或计算题: 把方法落实到步骤 | Diagram or calculation application

Question. Draw a PPC and show (i) unemployed resources becoming fully used and (ii) an improvement in technology affecting both products. Explain why the two changes are different. [6]

正确答案 | Correct answer. Label both axes and draw one PPC. For (i), move an inside point to a point on the same boundary: actual output rises because idle resources are used, but productive capacity is unchanged. For (ii), draw a second PPC outside the first and label the outward shift: improved technology raises the maximum attainable combinations.

图表推演 | Diagram analysis

PPC: movement and productive capacity



考试图表必须标出坐标轴、曲线名称、新旧均衡和变化方向。文字解释需说明原因、曲线移动以及价格或数量结果。

第 4 课 | 2.4 经济制度与资源配置 (Economic systems)

所属大章: 基本经济问题 | The basic economic problem

知识点精讲 | Core knowledge

Market economy 主要由价格和利润信号配置资源; planned economy

由政府决定主要生产什么、如何生产、为谁生产; 现实多是 mixed economy (混合经济)。评价制度不能只列优缺点: 市场激励创新但可能出现不平等、公共品不足和外部性; 政府可纠正失灵但也可能信息不足、官僚低效。

深度展开与真题化应用 | In-depth development (共 7 个知识点)

知识点 01/07 | 经济制度与资源配置: 三种制度回答同样的三个问题 | How three systems answer the same economic questions

所有经济体都必须决定 what to produce, how to produce and for whom to produce。市场经济主要依靠价格、利润和消费者选择; 计划经济主要依靠政府指令和公共所有; 混合经济同时使用市场与政府。判断制度时不要看“有没有政府”或“有没有企业”, 而要看关键资源配置决定由谁作出。

知识点 02/07 | 经济制度与资源配置: 比较激励、信息、效率与公平 | Comparing incentives, information, efficiency and equity

价格上涨向企业发出 scarcity 和 profitability

信号, 可能吸引资源进入该市场; 竞争鼓励降成本和创新。然而市场可能少供 public goods、忽略 external costs, 并依据购买力而非需要分配产品。政府可以提供医疗教育、征税或监管, 但也可能缺乏地方信息、反应缓慢或受到政治目标影响。现实结论通常取决于市场竞争程度、政府能力和产品性质。

知识点 03/07 | 经济制度与资源配置中的主体、变量与判断对象 | Agents, variables and object of analysis

Consumers、private firms 与 government 是三类关键决策者。市场制度主要通过 prices and profit 配置资源, 计划制度主要通过 state instructions, mixed economy 则按产品性质组合两种机制。

知识点 04/07 | 经济制度与资源配置的因果机制 | How the mechanism works

消费者对太阳能设备需求增加 → price/profit signal strengthens → firms enter and hire resources → supply expands → resources shift away from less profitable uses。计划经济则可能由政府根据能源目标直接配置资源。

知识点 05/07 | 经济制度与资源配置的关键英文表述 | Exam-ready explanation in English

A market system can process dispersed information through prices, while planning can direct resources towards social priorities. Neither system guarantees efficiency or fairness. A mixed economy attempts to combine market incentives with public provision and regulation.

知识点 06/07 | 经济制度与资源配置结论成立的条件 | When the conclusion may change

最佳组合取决于产品性质与政府能力。公共品、外部性强或信息不对称严重的领域通常需要更多干预, 普通消费品更适合竞争。

知识点 07/07 | 经济制度与资源配置的综合应用 | Integrated application

医疗由市场提供可带来选择与创新, 但 willingness and ability to pay 不等于 medical need。评价时把效率、信息不对称、外部收益、税收机会成本与政府执行能力放在同一结论中。

逐题精讲与评价拓展 | Exam workshop (共 2 项)

精讲 01/02 | 选择题: 区分最接近的概念 | Multiple-choice application

Question. Which feature is most characteristic of a market economy: consumer sovereignty, state production targets, equal income, or free public goods?

正确答案 | Correct answer. Consumer sovereignty, because purchases influence what profit-seeking firms produce.

精讲 02/02 | 结构题: 把情境发展成完整分析 | Structured application

Question. Discuss whether healthcare should be allocated entirely by the market.

参考答案 | Model answer. Market provision may encourage efficiency, choice and innovation. Competition can reduce cost. However, low-income patients may be unable to pay, consumers may lack medical information and healthcare creates external benefits. Public funding or regulation may improve access, although taxation has opportunity cost and government provision may be inefficient.

第 5 课 | 2.5 逐步判断：一个 PPC 题到底在问什么？

所属篇章：基本经济问题 | The basic economic problem

知识点精讲 | Core knowledge

Step 1 – 先识别位置。点在曲线内：不是“没有稀缺”，而是既定资源没有充分利用；可用 unemployment、inefficient organisation 或 recession 解释。点在曲线上：在当前技术下生产有效率，但不必然符合社会偏好。点在曲线外：当前不可行，除非贸易、技术或资源改变。

Step 2 – 再识别变化类型。“More workers receive training” 提高 labour productivity, PPC outward shift; “workers move from food to computers” 是 movement along PPC; “an earthquake destroys factories” 是 inward shift。答题句式：An improvement in human capital raises the productive potential of the economy, so the PPC shifts outwards.

Step 3 – 最后说代价。若由 A 转向 B，机会成本是少生产的 A。请避免写“B 的机会成本是 B”；机会成本必须是放弃的另一个选项。

深度展开与真题化应用 | In-depth development (共 6 个知识点)

知识点 01/06 | 逐步判断：一个 PPC 题到底在问什么：先识别题目在改变什么 | Identifying exactly what the question changes

PPC 题只有三类核心变化：点的位置、沿曲线移动、整条曲线移动。失业下降使生产点由曲线内向曲线靠近；军用品增加且消费品减少是沿同一曲线移动；技术、劳动力、资本存量或自然资源改变才移动整条曲线。题干出现“产量上升”并不自动说明 PPC 外移，因为利用闲置资源也能提高实际产量。

知识点 02/06 | 逐步判断：一个 PPC 题到底在问什么：用反事实检查答案 | Checking a PPC conclusion with a counterfactual

问自己：资源和技术是否改变？若没有，就不能画新 PPC。再问：两种产品是否一增一减？若是，多半是沿曲线移动。若两种最大产量都提高，才通常整体外移。最后区分 short-run actual output 与 long-run productive potential。

知识点 03/06 | 逐步判断：一个 PPC 题到底在问什么中的主体、变量与判断对象 | Agents, variables and object of analysis

题干出现 reallocate resources 时分析沿曲线移动；出现 unemployment falls 时分析曲线内点向边界移动；出现 technology、education、migration 或 disaster 时才判断 PPC 是否外移、内移或偏向性旋转。

知识点 04/06 | 逐步判断：一个 PPC 题到底在问什么的数字或现实情境 | Numerical or real-world application

旧 PPC 最大生产 50 台机器或 100 吨粮食；技术只改善机器生产，使最大机器升至 80、粮食仍为 100。图形绕粮食截距向外旋转，而不是平行移动。

知识点 05/06 | 逐步判断：一个 PPC 题到底在问什么的图表与数据读法 | Diagram and data interpretation

先读两条坐标轴代表的产品，再判断点在曲线上、曲线内还是曲线外。沿同一条 PPC 移动表示产品组合改变；资源数量、资源质量或技术改变才使整条 PPC 移动。图形结论必须同时写出 gained output 与 forgone output。

知识点 06/06 | 逐步判断：一个 PPC 题到底在问什么最容易混淆的口径 | Common misconception

“Output rises” 不必然等于 “economic growth”。使用闲置资源提高 actual output 与 productive potential 增长必须分开。

逐题精讲与评价拓展 | Exam workshop (共 3 项)

精讲 01/03 | 选择题: 区分最接近的概念 | Multiple-choice application

Question. What happens to the PPC when unemployed workers find jobs but the labour force and productivity are unchanged?

正确答案 | Correct answer. The production point moves towards the existing PPC. Productive potential is unchanged.

精讲 02/03 | 结构题: 把情境发展成完整分析 | Structured application

Question. Analyse the PPC effects of a government transferring resources from consumer goods to capital goods.

参考答案 | Model answer. In the current period, the economy moves along its PPC towards more capital goods and fewer consumer goods. The sacrificed consumer goods are the opportunity cost. If capital formation raises future productivity, the future PPC may shift outwards, allowing more of both goods.

精讲 03/03 | 图表或计算题: 把方法落实到步骤 | Diagram or calculation application

Question. An economy moves from a point inside its PPC to a point on the curve. Does this prove that its resources have increased? Explain. [3]

正确答案 | Correct answer. No. The movement can result from lower unemployment or better use of existing resources. Actual production rises toward the existing maximum, but the PPC itself shifts only if the quantity or quality of resources, productivity or technology changes.

图表推演 | Diagram analysis

PPC: movement and productive capacity



考试图表必须标出坐标轴、曲线名称、新旧均衡和变化方向。文字解释需说明原因、曲线移动以及价格或数量结果。

Unit 2 | 资源配置

The allocation of resources

单元导读 | Unit introduction

本单元把买方和卖方的决定放进同一张图。从 demand 与 supply 的准确定义开始，逐步处理 movement、shift、market equilibrium、price mechanism、elasticity 与 market failure。每一条曲线变化都要回答四个问题：谁的行为变了、哪个非价格因素变了、哪条曲线向哪边移动、均衡价格和数量如何变化。

学习主线 | Learning route

阅读供需图时先固定其他条件，再区分 own-price change 与 non-price determinant。涉及两条曲线同时移动时，分别判断每次移动对 P 与 Q 的作用，再合并；若方向冲突，结论必须写 depends on the relative size of the shifts。

English overview

Demand and supply are plans, not completed transactions. A change in the product's own price causes movement along a curve; a non-price factor shifts the whole curve. The price mechanism uses shortages and surpluses to move a market towards equilibrium, while elasticity measures the size of the response.

本单元课时 | Lessons in this unit

第 6 课 | 2.1 市场、买方与卖方 | Market, buyers and sellers; 第 7 课 | 2.2 需求、个人需求与市场需求 | Individual and market demand; 第 8 课 | 2.3 需求量移动与需求曲线移位 | Movement along and shift of demand; 第 9 课 | 2.4 供给与供给决定因素 | Supply and its determinants; 第 10 课 | 2.5 价格机制、均衡、短缺与过剩 | Price mechanism and equilibrium; 第 11 课 | 2.6 需求和供给同时移动 | Simultaneous changes; 第 12 课 | 2.7 PED、PES 与销售收入 | Elasticity and revenue; 第 13 课 | 2.8 市场失灵与政府干预 | Market failure and intervention

第 6 课 | 2.1 市场、买方与卖方 | Market, buyers and sellers

所属大章: 资源配置 | The allocation of resources

知识点精讲 | Core knowledge

市场 (market) 是使买者与卖者相互接触、进行交换的一种安排; 它可以是实体市场, 也可以是网络平台。买方通常是 consumers, 他们购买商品与服务满足需要; 卖方通常是 firms, 他们组织生产要素并提供产品。市场不要求买卖双方在同一地点。

English explanation. A market is an arrangement that brings buyers and sellers into contact. Buyers express demand; sellers express supply. Their decisions interact through prices. In Cambridge questions, identify the product market first and distinguish a consumer from a producer.

深度展开与真题化应用 | In-depth development (共 4 个知识点)

知识点 01/04 | 市场、买方与卖方: 市场是一种安排 | A market is an arrangement, not only a place

市场不一定有店铺或固定地点。网约车平台、外汇市场和劳动力市场都让潜在买者与卖者交换。买者形成 demand, 卖者形成 supply, 价格承担 rationing、signalling 和 incentive 功能。分析前要先界定市场: 讨论“饮料市场”与“无糖饮料市场”可能得到不同的竞争者和替代品。

知识点 02/04 | 市场、买方与卖方: 市场参与者的目标与约束 | Objectives and constraints of buyers and sellers

消费者通常希望在预算内提高 satisfaction; 企业希望实现利润或其他目标, 但受成本、技术和竞争限制。政府也可能作为买方采购疫苗, 或作为卖方提供交通服务。市场中的“买方”和“卖方”是经济角色, 不是固定身份; 家庭在产品市场是买方, 在劳动力市场却出售 labour services。

知识点 03/04 | 市场、买方与卖方中的主体、变量与判断对象 | Agents, variables and object of analysis

Buyers 形成 demand, sellers 形成 supply; 市场可以是实体地点, 也可以是线上或其他使双方接触的安排。价格、数量与交易时期必须明确, 才构成可比较的市场信息。

知识点 04/04 | 市场、买方与卖方的关键英文表述 | Exam-ready explanation in English

A market includes potential transactions, not only completed sales. Its boundaries matter because a narrow definition may show few competitors, while a broad definition may include many substitutes. Online interaction is sufficient to bring buyers and sellers into contact.

逐题精讲与评价拓展 | Exam workshop (共 2 项)

精讲 01/02 | 选择题: 区分最接近的概念 | Multiple-choice application

Question. Which is a market: only a physical shop, only a stock exchange, any arrangement connecting buyers and sellers, or any government office?

正确答案 | Correct answer. Any arrangement connecting buyers and sellers. Physical location is not required.

精讲 02/02 | 结构题: 把情境发展成完整分析 | Structured application

Question. Explain two ways prices influence buyers and sellers in a market.

参考答案 | Model answer. A higher price rations the product by reducing the quantity consumers are willing and able to buy. It also provides sellers with an incentive to increase quantity supplied because potential revenue and profit rise. These responses help coordinate the two sides of the market.

第 7 课 | 2.2 需求、个人需求与市场需求 | Individual and market demand

所属大章: 资源配置 | The allocation of resources

知识点精讲 | Core knowledge

需求 (demand) 是消费者在一定时期、不同价格下愿意并且能够购买的数量。个人需求是一个消费者的需求; 市场需求是在每一个价格水平把所有消费者的需求量横向相加。若价格为 \$10, A 购买 500 单位, B 购买 400 单位, 则市场需求量为 900。

English explanation. Demand is the willingness and ability to buy a product at different prices over a period of time. Market demand is the horizontal sum of all individual demand: quantities, not prices, are added at each price.

深度展开与真题化应用 | In-depth development (共 6 个知识点)

知识点 01/06 | 需求、个人需求与市场需求: Demand 包含 willing、able、price 与 time | The four elements of an accurate demand definition

想买但付不起不是有效需求, 有能力但不愿买也不是需求。需求表列出一定时期内不同价格对应的 quantity demanded, 曲线只是把表格图形化。需求定律说明 ceteris paribus 下价格越低, 需求量通常越大, 因为替代效应和收入效应共同作用。

知识点 02/06 | 需求、个人需求与市场需求: 市场需求为何是横向加总 | Why market demand is a horizontal sum

同一个价格下, 把每位消费者愿意购买的数量相加。例如 \$10 时 A 买 5 个、B 买 4 个, market quantity demanded 为 9 个; 不能把两个 \$10 相加为 \$20。若第三位消费者在该价位愿意买 2 个, 市场需求量变成 11, 整条市场需求曲线也可能向右。

知识点 03/06 | 需求、个人需求与市场需求中的主体、变量与判断对象 | Agents, variables and object of analysis

Individual demand 的主体是一名消费者; market demand 的主体是同一市场内全部消费者。纵轴价格保持相同, 横轴数量逐人相加, 单位与时期也必须一致。

知识点 04/06 | 需求、个人需求与市场需求的数字或现实情境 | Numerical or real-world application

价格 \$12 时 A 买 3、B 买 5、C 买 0, 市场需求量 8; 价格 \$8 时分别买 5、8、2, 市场需求量 15。每行加数量, 不能加价格。

知识点 05/06 | 需求、个人需求与市场需求的图表与数据读法 | Diagram and data interpretation

在每一个完全相同的价格上, 把所有消费者的 quantity demanded 横向相加。价格不能相加, 曲线斜率也不能直接相加; 市场曲线上的每一点都来自同价位的数量总和。

知识点 06/06 | 需求、个人需求与市场需求的英文表述 | Exam-ready explanation in English

Willingness without purchasing power is a desire, not effective demand. Ability without willingness also produces no demand. The time period matters: ten units per day and ten units per year describe very different demand.

逐题精讲与评价拓展 | Exam workshop (共 3 项)

精讲 01/03 | 选择题: 区分最接近的概念 | Multiple-choice application

Question. At a price of \$5, three consumers demand 4, 7 and 9 units. What is market quantity demanded?

正确答案 | Correct answer. 20 units. Market demand is the horizontal sum of individual quantities at the same price.

精讲 02/03 | 结构题: 把情境发展成完整分析 | Structured application

Question. Explain why a demand curve usually slopes downwards.

参考答案 | Model answer. When price falls, the product becomes cheaper relative to substitutes, so consumers may switch towards it. The lower price also increases real purchasing power, enabling consumers to buy more. Hence quantity demanded rises, *ceteris paribus*.

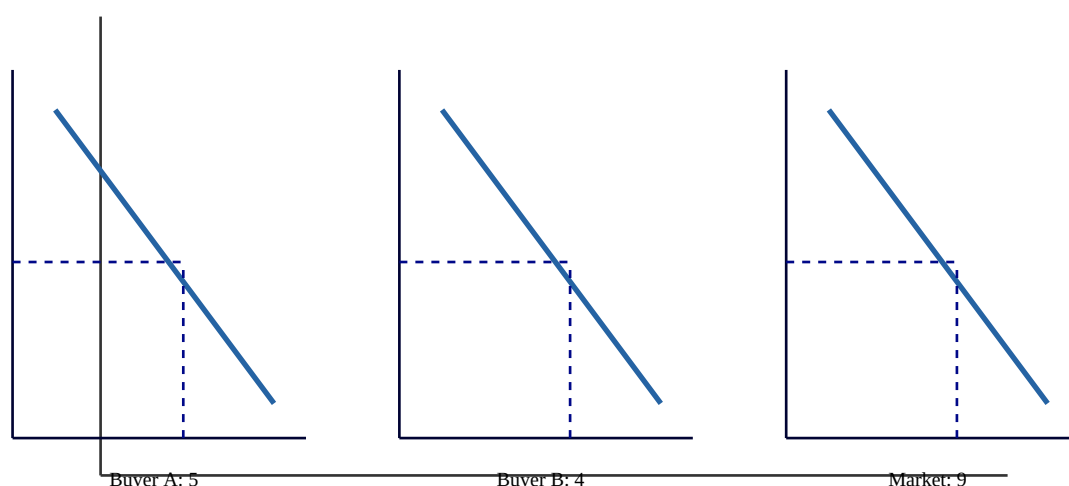
精讲 03/03 | 图表或计算题: 把方法落实到步骤 | Diagram or calculation application

Question. At a price of \$8, Ana demands 3 units, Ben 5 and Chen 4. Calculate market demand and state what must be held constant when adding the figures. [3]

正确答案 | Correct answer. Market quantity demanded is $3 + 5 + 4 = 12$ units. The price, time period and product unit must be the same for every consumer; prices are not added.

图表推演 | Diagram analysis

Market demand = sum of individual demand



考试图表必须标出坐标轴、曲线名称、新旧均衡和变化方向。文字解释需说明原因、曲线移动以及价格或数量结果。

第 8 课 | 2.3 需求量移动与需求曲线移位 | Movement along and shift of demand

所属大章: 资源配置 | The allocation of resources

知识点精讲 | Core knowledge

商品自身价格下降会导致 extension in demand / increase in quantity demanded, 表现为沿同一条需求曲线向右下移动; 自身价格上升会导致 contraction。收入、相关商品价格、广告、偏好、人口结构、天气和未来价格预期属于非价格决定因素, 它们改变整条需求曲线。正常品收入增加使需求右移; 劣等品收入增加可能使需求左移。替代品价格上涨会增加本商品需求; 互补品价格上涨会减少本商品需求。

English explanation. A change in the product's own price causes a movement along the existing demand curve. A non-price determinant causes the whole curve to shift. Write "quantity demanded increases" for an extension and "demand increases" for a rightward shift.

深度展开与真题化应用 | In-depth development (共 7 个知识点)

知识点 01/07 | 需求量移动与需求曲线移位: 一条曲线还是两条曲线 | One demand curve or a shifted curve

判断的第一问永远是: 变的是商品自己的价格, 还是其他条件? Own price change 只改变 quantity demanded, 点沿原曲线移动; non-price determinant 改变 demand, 整条曲线移动。价格下降形成 extension in demand, 价格上涨形成 contraction in demand。收入、偏好、广告、人口、预期和相关商品价格不在纵轴上, 所以需要画新曲线。

知识点 02/07 | 需求量移动与需求曲线移位: 相关商品与收入的方向判断 | Income and related-goods effects on demand

咖啡与茶是 substitutes: 咖啡价格上升, 部分消费者改买茶, 茶的 demand increases。汽车与汽油是 complements: 汽油价格上升使驾车总成本增加, 汽车需求下降。收入上升令 normal good 需求增加, 却可能令 inferior good 需求减少。先识别关系, 再写消费者选择改变, 最后判断 $D_0 \rightarrow D_1$ 的方向。

知识点 03/07 | 需求量移动与需求曲线移位中的主体、变量与判断对象 | Agents, variables and object of analysis

Own price 位于需求图纵轴, 因此其变化表现为同一 D 上的点移动。Income、taste、population、expectations 及 related goods prices 不在坐标轴上, 因此改变每一价格下的购买计划, 使整条 D 移位。

知识点 04/07 | 需求量移动与需求曲线移位的因果机制 | How the mechanism works

tea price rises $\rightarrow$ coffee is a substitute $\rightarrow$ consumers switch from tea to coffee $\rightarrow$ coffee demand increases $\rightarrow$ coffee demand curve shifts right $\rightarrow$ coffee equilibrium price and quantity tend to rise.

知识点 05/07 | 需求量移动与需求曲线移位的图表与数据读法 | Diagram and data interpretation

商品自身价格改变时, 标出同一条 D 曲线上的 A、B 两点; 收入、偏好、人口、预期或相关商品价格改变时, 才画 D_0 到 D_1 。作答时必须用 movement along 与 shift 的不同术语。

知识点 06/07 | 需求量移动与需求曲线移位的关键英文表述 | Exam-ready explanation in English

The curve represents the relationship between the product's own price and quantity demanded. Therefore, a change in any other determinant changes the relationship itself and requires a new curve. State both the determinant and why willingness or ability to buy changes.

知识点 07/07 | 需求量移动与需求曲线移位最容易混淆的口径 | Common misconception

“价格下降使需求增加”在 Cambridge 精确口径中应写 quantity demanded increases 或 extension in demand。

逐题精讲与评价拓展 | Exam workshop (共 3 项)

精讲 01/03 | 选择题: 区分最接近的概念 | Multiple-choice application

Question. The price of petrol rises. What happens to demand for petrol cars, assuming they are complements?

正确答案 | Correct answer. Demand for petrol cars decreases and its demand curve shifts left. The car price need not have changed.

精讲 02/03 | 结构题: 把情境发展成完整分析 | Structured application

Question. Analyse how higher household income may affect demand for bus travel.

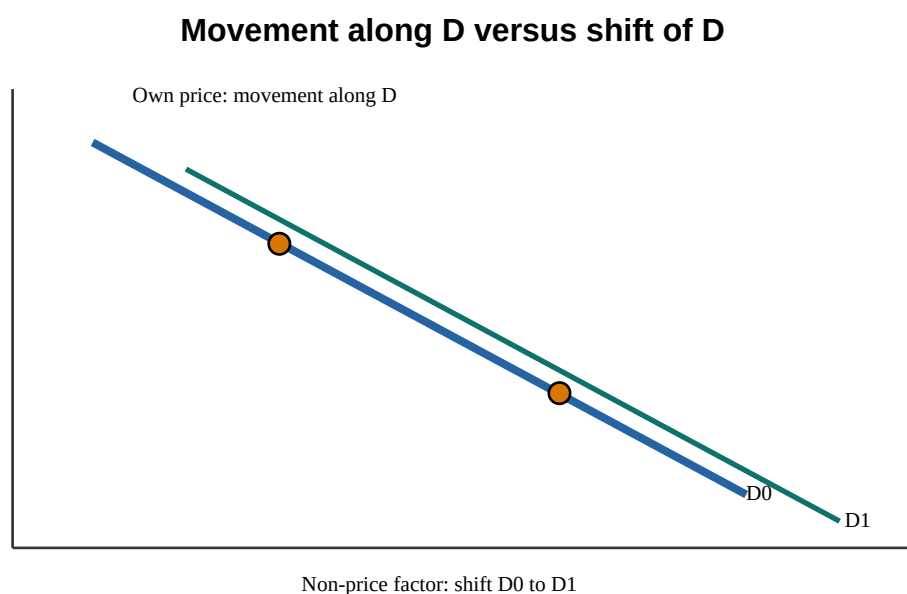
参考答案 | Model answer. If bus travel is an inferior service, higher income enables households to switch to cars or taxis. Demand for bus travel falls and its curve shifts left. If buses are high-quality and environmentally preferred, they may instead be normal, so the result depends on consumer behaviour.

精讲 03/03 | 图表或计算题: 把方法落实到步骤 | Diagram or calculation application

Question. Draw diagrams to distinguish a fall in the price of tea from a rise in the price of coffee, a substitute for tea. [6]

正确答案 | Correct answer. For the fall in tea price, use one tea demand curve and show a downward movement to a larger quantity demanded. For the rise in coffee price, keep tea price unchanged and shift the whole tea demand curve right because consumers switch from coffee to tea.

图表推演 | Diagram analysis



考试图表必须标出坐标轴、曲线名称、新旧均衡和变化方向。文字解释需说明原因、曲线移动以及价格或数量结果。

第 9 课 | 2.4 供给与供给决定因素 | Supply and its determinants

所属篇章：资源配置 | The allocation of resources

知识点精讲 | Core knowledge

供给 (supply) 是在一定时期、不同价格下厂商愿意并且能够出售的数量。商品自身价格变化造成沿供给曲线移动；生产成本、技术、间接税、补贴、其他产品利润、企业数量、天气、战争和原材料发现等会使整条供给曲线移动。成本下降或技术改善使供给右移；间接税提高单位成本，使供给左移；补贴使供给右移。

English explanation. Supply is the willingness and ability of producers to sell a product at different prices over a period of time. A rise in the product's own price causes an extension in supply. Lower costs, better technology or a subsidy increase supply.

深度展开与真题化应用 | In-depth development (共 7 个知识点)

知识点 01/07 | 供给与供给决定因素：Supply 看的是厂商计划销售量 | Supply as producers planned sales

库存很多不等于市场供给一定多；供给指厂商在不同价格下愿意且能够出售的数量。价格上涨通常提高边际收益，使扩产更有吸引力，形成 extension in supply。若生产能力、成本或政策改变，则每个价格下都可能愿意出售更多或更少，曲线才发生 shift。

知识点 02/07 | 供给与供给决定因素：决定因素的机制不能只背方向 | How costs, technology and policy shift supply

工资、能源和原材料价格下降会降低 unit cost，提高每单位利润，因此 supply increases。技术改善提高 productivity，也会降低 average cost。Indirect tax 提高单位成本，使 S 左移；subsidy 降低有效成本，使 S 右移。好天气提高农作物产量；灾害、战争或资源枯竭限制生产。其他产品更有利可图时，企业可能转移资源，当前产品供给减少。

知识点 03/07 | 供给与供给决定因素中的主体、变量与判断对象 | Agents, variables and object of analysis

Supply 的主体是 producers，核心是不同价格下 planned quantity supplied。Own price 改变利润边际并造成沿 S 移动；input costs、technology、tax、subsidy、weather 与 number of firms 改变整条 S。

知识点 04/07 | 供给与供给决定因素的因果机制 | How the mechanism works

electricity cost falls → unit production cost falls → profit per unit at every price rises → firms are willing and able to sell more → supply shifts right → equilibrium price tends to fall and quantity rises.

知识点 05/07 | 供给与供给决定因素的图表与数据读法 | Diagram and data interpretation

纵轴是 price，横轴是 quantity。先找 demand 与 supply 的交点，再从价格线分别读 Qd 和 Qs。曲线移动时先说明是哪一个非价格因素改变，再比较新旧 equilibrium price 与 equilibrium quantity。

知识点 06/07 | 供给与供给决定因素的关键英文表述 | Exam-ready explanation in English

Supply describes planned sales, not all existing stock. A warehouse may contain goods that the owner is unwilling to sell at the current price. Explain shifts through costs, profitability and productive capacity rather than memorising directions only.

知识点 07/07 | 供给与供给决定因素最容易混淆的口径 | Common misconception

“供给增加”不能仅由产品自身价格上涨造成；那是 quantity supplied 增加。

逐题精讲与评价拓展 | Exam workshop (共 3 项)

精讲 01/03 | 选择题: 区分最接近的概念 | Multiple-choice application

Question. Which change shifts the supply of bread left: lower flour prices, a subsidy, improved ovens, or a tax on each loaf?

正确答案 | Correct answer. A tax on each loaf, because it raises unit cost and reduces profitability at every market price.

精讲 02/03 | 结构题: 把情境发展成完整分析 | Structured application

Question. Explain two reasons why the supply of agricultural products may be price inelastic in the short run.

参考答案 | Model answer. Crops require time to grow, so farmers cannot quickly raise current harvests when price rises. Available land and storage may also be limited. Hence the percentage change in quantity supplied is smaller than the percentage change in price.

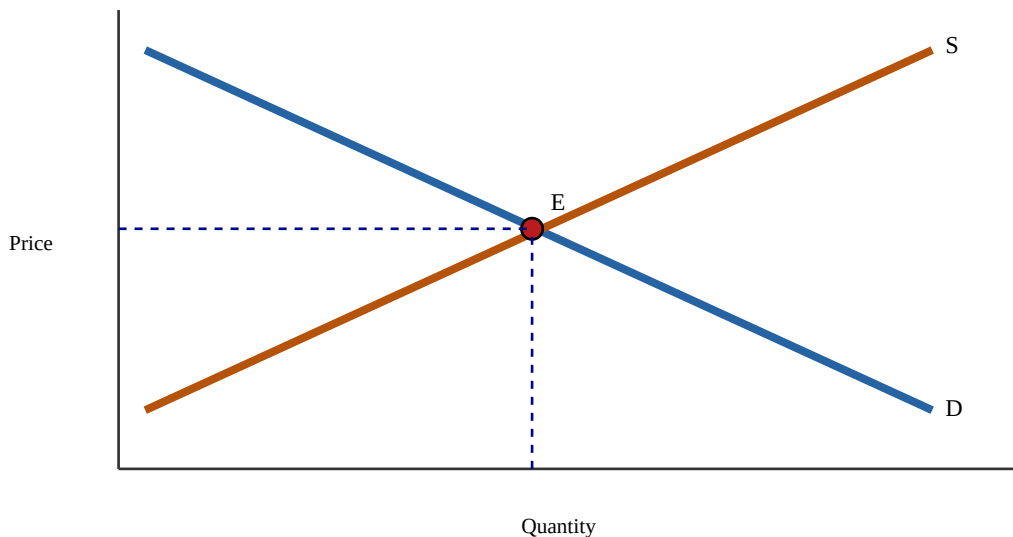
精讲 03/03 | 图表或计算题: 把方法落实到步骤 | Diagram or calculation application

Question. Draw and explain the effect of a subsidy paid to bicycle producers. [4]

正确答案 | Correct answer. Label Price and Quantity, draw D and S1, then shift supply right to S2. A subsidy lowers the effective cost of production and increases the profit obtainable at each market price, so producers are willing and able to supply more.

图表推演 | Diagram analysis

Demand, supply and equilibrium



考试图表必须标出坐标轴、曲线名称、新旧均衡和变化方向。文字解释需说明原因、曲线移动以及价格或数量结果。

第 10 课 | 2.5 价格机制、均衡、短缺与过剩 | Price mechanism and equilibrium

所属大章: 资源配置 | The allocation of resources

知识点精讲 | Core knowledge

价格机制通过价格变化协调消费者和企业的决定，并回答生产什么、如何生产和为谁生产。均衡出现在 quantity demanded equals quantity supplied。低于均衡价时出现 shortage，买方竞争产生价格上升压力；高于均衡价时出现 surplus，库存增加产生价格下降压力。

English explanation. The price mechanism allocates resources through changes in demand, supply and price. A shortage causes upward pressure on price; a surplus causes downward pressure. The market moves towards equilibrium where planned purchases equal planned sales.

深度展开与真题化应用 | In-depth development (共 10 个知识点)

知识点 01/10 | 价格机制到底是什么 | How the price mechanism coordinates a market

价格机制 (price mechanism) 是价格根据 demand 与 supply 的变化而调整，并据此配置稀缺资源的过程。它不是某个人发出命令，而是大量消费者与企业分别作出决定后，通过价格相互协调。完整分析包含三层：外部事件先改变 demand 或 supply；市场随后出现 shortage 或 surplus；价格再发生变化，使 quantity demanded 和 quantity supplied 沿原曲线调整，市场趋向新均衡。

例如某城市突然流行电动自行车，需求曲线右移。在原价格下 Q_d 大于 Q_s ，门店很快售罄。短缺向卖方传递“消费者愿意购买更多”的信息，价格可能上涨，企业也更愿意进货或扩产，最终形成较高的 equilibrium price 与较大的 equilibrium quantity。这条链比“需求增加，所以价格上涨”更完整。

知识点 02/10 | 信号功能 | Signalling function

价格像一个不断更新的信息系统。价格上涨可能表示商品变得相对稀缺，原因可能是需求增加或供给减少；价格下降可能表示需求减弱或供给增加。企业不需要知道每一位消费者为什么改变选择，也能从销量、库存和价格观察市场信号。不过，单看价格不能确定曲线移动原因：同样是 price rise，可能来自 D right，也可能来自 S left，两者对 equilibrium quantity 的影响相反。

English explanation. The signalling function means that changes in price communicate information about relative scarcity. A rising price may signal stronger demand or reduced supply. Producers should also consider whether the change is temporary, the cost of expansion and expected future demand.

知识点 03/10 | 激励功能 | Incentive function

价格上升通常提高每单位潜在 revenue 和 profit，形成增产激励。现有企业可能增加 workers' hours、使用 spare capacity 或购买 machinery；较长时期内，新企业可能 enter the market。于是 quantity supplied 先沿供给曲线增加，若企业数量或生产能力增长，整条 supply curve 还可能右移。持续低价和亏损则促使企业减产或退出，让 labour、capital 和 land 转向回报更高的用途。

激励并不保证供给立刻大幅增加。鲜花、农作物和住房需要生产时间，PES 较低；制造商若有库存和闲置设备，反应可能更快。高价也可能由间接税造成，并不表示生产者实际收入同比上涨。因此评价要考虑 time period、PES、capacity 和 costs。

English explanation. The incentive function changes behaviour. A higher market price can raise expected profit, encouraging existing firms to expand and new firms to enter. The speed and size of the response depend on price elasticity of supply, spare capacity, stocks and the production period.

知识点 04/10 | 配给功能 | Rationing function

当商品数量有限，价格上涨会减少 quantity demanded，把产品分配给 willing and able to pay 的消费者。演唱会座位、城市中心住房或干旱时期的水都可能通过价格配给。这里的 able to pay 很重要：市场配给依据购买力，不是依据“谁最需要”。所以价格机制可能有效消除 shortage，却产生 equity 问题；低收入家庭可能买不起必需品。政府可使用 subsidy、direct provision 或 maximum price，但这些政策也可能造成财政成本或新的短缺。

English explanation. The rationing function allocates a scarce product among competing consumers. A higher price reduces quantity demanded, so the available supply goes to those willing and able to pay. This may restore equilibrium, but it does not necessarily produce a fair distribution.

知识点 05/10 | 价格机制怎样回答三个基本经济问题 | How prices answer the three economic questions

生产什么 | What to produce? 消费者通过 demand

表达偏好。需求增加通常使价格与利润上升，企业把资源转入该市场；需求下降则使资源退出。准确说法是 consumer sovereignty works through demand, price and profit signals。

如何生产 | How to produce? 企业比较 labour、capital、land

等要素的相对价格和生产率，选择成本较低且符合质量要求的方法。若工资相对机器价格上升，企业可能采用 capital-intensive production；但机器购置成本、技能、产量规模和质量也会影响决定，所以不能简单写“永远选最便宜要素”。

为谁生产 | For whom to produce? 商品主要流向愿意且能够支付市场价格的人，购买力来自

wages、rent、interest、profit、wealth 或政府转移。市场按 income and purchasing power 分配，而不是按 need 分配，这正是效率与公平之间的重要张力。

English summary. Consumer demand and profit signals influence what is produced. Relative factor prices and productivity influence how it is produced. Income and purchasing power influence for whom it is produced. These outcomes may be efficient, but they are not necessarily equitable.

知识点 06/10 | 均衡是一组价格与数量 | Equilibrium price and equilibrium quantity

均衡点 E 同时给出 equilibrium price P_e 和 equilibrium quantity Q_e 。 P_e 以下 $Q_d > Q_s$ ，出现 shortage； P_e 以上 $Q_s > Q_d$ ，出现 surplus。价格变化使需求量和供给量沿各自曲线移动，直到 planned purchases equal planned sales。均衡不是“公平价格”，也不保证每个人满意；它只表示市场没有内在的价格变化压力。

知识点 07/10 | 供过于求 | Surplus: $Q_s > Q_d$

假设 equilibrium price 是 \$6、quantity 是 100。若价格为 \$8，消费者计划购买 70，企业计划出售 130，因此 $\text{surplus} = Q_s - Q_d = 60$ 。库存积压使企业降价；价格下降造成 extension in demand 和 contraction in supply，差额逐渐缩小，直到回到均衡。D 与 S 都没有移动，因为引起调整的是商品自身价格。

English chain. Price is above equilibrium $\rightarrow$ quantity supplied exceeds quantity demanded $\rightarrow$ unsold stocks accumulate $\rightarrow$ sellers reduce price $\rightarrow$ quantity demanded extends and quantity supplied contracts $\rightarrow$ the surplus disappears at equilibrium.

口诀 $Q_s > Q_d = \text{surplus}$ 只负责快速判断；完整答案还要写 unsold stock 和 downward pressure on price。

知识点 08/10 | 供不应求 | Shortage: $Q_d > Q_s$

若价格为 \$4，消费者计划购买 140，企业只计划出售 60，因此

$\text{shortage} = Q_d - Q_s = 80$ 。消费者竞争、排队或缺货使价格承受上升压力；价格上涨造成 contraction in demand 和 extension in supply，直到 $Q_d = Q_s$ 。若 maximum price 阻止价格上升，shortage 可能持续，并出现 waiting lists、black markets 或 non-price rationing。

English chain. Price is below equilibrium $\rightarrow$ quantity demanded exceeds quantity supplied $\rightarrow$ excess demand creates upward pressure on price $\rightarrow$ quantity demanded contracts and quantity supplied extends $\rightarrow$

equilibrium is restored.

口诀 $Q_d > Q_s = \text{shortage}$ 。Shortage 不是“供给减少”的同义词，而是某一个价格下 Q_d 与 Q_s 的差额。

知识点 09/10 | 用供需数据验证两张图 | Verifying surplus and shortage with data

Price \$8: Q_d 70, Q_s 130 $\rightarrow$ surplus 60 $\rightarrow$ downward pressure on price.

Price \$6: Q_d 100, Q_s 100 $\rightarrow$ equilibrium $\rightarrow$ no tendency for price to change.

Price \$4: Q_d 140, Q_s 60 $\rightarrow$ shortage 80 $\rightarrow$ upward pressure on price.

读表顺序：在同一价格比较 Q_d 和 Q_s ；用较大者减较小者计算

gap；最后判断价格方向。不要用不同价格的两行数据相减，也不要写 shortage 写成负数。

知识点 10/10 | 图表题的得分细节 | Required labels and adjustment arrows

两张图都标 Price 纵轴、Quantity 横轴、向下的 D、向上的 S、交点 E 和均衡价 P_e 。Surplus 图在 P_e 上方画

P1：左边是 Q_d 、右边是 Q_s ，中间标 surplus。Shortage 图在 P_e 下方画 P1：左边是 Q_s 、右边是 Q_d ，中间标 shortage。只画水平线而不标 Q_d/Q_s ，阅卷人无法确认你是否理解差额。

逐题精讲与评价拓展 | Exam workshop (共 8 项)

精讲 01/08 | 选择题 1: 识别过剩 | MCQ 1: identifying a surplus

Question. At the current price, firms plan to sell 1 200 units while consumers plan to buy 850 units. Which outcome is most likely in a free market?

A Demand increases. B Price falls. C Supply shifts left. D Price rises.

正确答案 | Correct answer: B – Price falls. Q_s exceeds Q_d by 350 units, so unsold stocks accumulate. Sellers have an incentive to reduce price. The price fall causes an extension in quantity demanded and a contraction in quantity supplied; neither curve has to shift.

精讲 02/08 | 选择题 2: 识别短缺 | MCQ 2: identifying a shortage

Question. The market price is below equilibrium. Which sequence correctly describes the adjustment?

A surplus $\rightarrow$ price rises; B shortage $\rightarrow$ price rises; C shortage $\rightarrow$ demand shifts right; D surplus $\rightarrow$ supply shifts left.

正确答案 | Correct answer: B – shortage $\rightarrow$ price rises. Below equilibrium, $Q_d > Q_s$. Competition among buyers produces upward pressure on price. Higher price contracts quantity demanded and extends quantity supplied until $Q_d = Q_s$.

精讲 03/08 | 数据题 3: 计算缺口 | Data question 3: calculating the gap

Question. At \$5, Q_d is 160 and Q_s is 90. At \$8, Q_d is 110 and Q_s is 140. Identify and calculate the disequilibrium at each price.

正确答案 | Correct answer. At \$5 there is a shortage of 70 units because $160 - 90 = 70$. At \$8 there is a surplus of 30 units because $140 - 110 = 30$. The \$5 price faces upward pressure; the \$8 price faces downward pressure.

精讲 04/08 | 结构题 4: 短缺如何被消除 | Structured question 4: removing a shortage

Question. Explain how the price mechanism may remove a shortage of bottled water. [4]

参考答案 | Model answer. A shortage means quantity demanded is greater than quantity supplied at the current price [1]. Buyers compete for limited bottles, creating upward pressure on price [1]. The higher price

causes a contraction in quantity demanded and an extension in quantity supplied [1]. Adjustment continues until Q_d equals Q_s at the equilibrium price [1].

精讲 05/08 | 结构题 5: 价格机制的三项功能 | Structured question 5: three functions

Question. Analyse how a rise in the price of solar panels can perform the signalling, incentive and rationing functions. [6]

参考答案 | Model answer. The higher price signals that panels have become relatively scarce or that demand has strengthened [1]. It tells firms that potential revenue and profit may be higher [1]. This gives existing producers an incentive to expand and may attract new firms, increasing supply over time [1–2]. The higher price also reduces quantity demanded [1], rationing the limited panels to consumers willing and able to pay [1].

精讲 06/08 | 评价题 6: 价格是否总能有效配置资源 | Evaluation question 6

Question. Discuss whether the price mechanism will always allocate essential medicines efficiently. [8]

参考答案 | Model answer. Prices can signal scarcity, encourage production and ration limited stocks. Profit can attract investment and innovation. However, willingness and ability to pay may not reflect medical need; low-income patients can be excluded. Patents, imperfect information and market power may keep prices high, while positive external benefits from treatment may cause under-consumption. A justified conclusion is that markets can coordinate ordinary supply, but subsidy, regulation or direct provision may be needed for essential medicines. The best policy depends on fiscal cost, targeting and government information.

精讲 07/08 | 图表题 7: 作图与正确结论 | Diagram question 7

Question. Draw a demand and supply diagram showing a price above equilibrium and explain the adjustment.

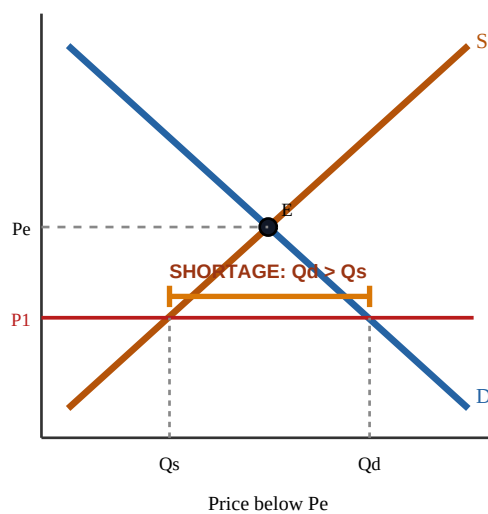
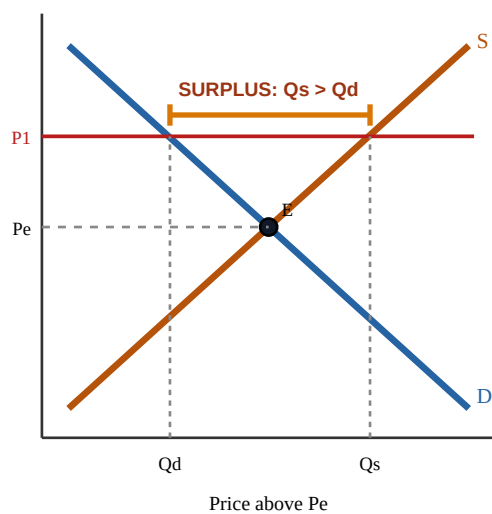
正确答案 | Correct answer. The diagram must label Price, Quantity, D , S , E , P_e and a higher P_1 . At P_1 , mark Q_d on the left and Q_s on the right; the horizontal gap is surplus. State $Q_s > Q_d \rightarrow$ surplus $\rightarrow$ unsold stocks $\rightarrow$ downward pressure on price $\rightarrow$ extension in demand and contraction in supply $\rightarrow$ equilibrium.

精讲 08/08 | 评价题 8: 价格调整何时受阻 | Evaluation question 8

Question. Explain two reasons why a shortage may persist even when there is upward pressure on price. [4]

参考答案 | Model answer. A government may impose a maximum price below equilibrium, preventing the legal price from rising; quantity demanded therefore remains above quantity supplied. Alternatively, long contracts or very low short-run PES may delay producers' response, so supply cannot increase quickly. Queues, rationing or a black market may then appear. Each reason must identify the obstacle and connect it to the continuing gap between Q_d and Q_s .

Surplus and shortage around equilibrium



考试图表必须标出坐标轴、曲线名称、新旧均衡和变化方向。文字解释需说明原因、曲线移动以及价格或数量结果。

第 11 课 | 2.6 需求和供给同时移动 | Simultaneous changes

所属大章: 资源配置 | The allocation of resources

知识点精讲 | Core knowledge

当需求和供给同时右移时，均衡数量一定增加，但均衡价格方向取决于两条曲线移动的相对幅度。如果需求增加幅度大于供给增加，价格上升；若供给增加幅度更大，价格下降。若一条右移、一条左移，则可能只能确定价格或数量中的一个。

English explanation. When both demand and supply increase, equilibrium quantity rises, but the effect on equilibrium price is uncertain. It depends on the relative size of the two shifts. State what is certain and what is indeterminate.

深度展开与真题化应用 | In-depth development (共 6 个知识点)

知识点 01/06 | 需求和供给同时移动: 先单独判断，再合并 | Combining separate demand and supply effects

同时移动题不能凭直觉直接下结论。先只看 demand shift 对 P、Q 的作用，再只看 supply shift，最后比较方向。D ↑ 使 P ↑ Q ↑；S ↑ 使 P ↓ Q ↑，合并后 Q 一定 ↑，P 不确定。D ↓ 与 S ↓ 使 Q 一定 ↓，P 不确定。D ↑ 与 S ↓ 使 P 一定 ↑，Q 不确定；D ↓ 与 S ↑ 使 P 一定 ↓，Q 不确定。

知识点 02/06 | 需求和供给同时移动: “不确定”也需要经济解释 | Explaining an indeterminate price or quantity

不确定不是不会做，而是缺少 relative magnitude。若 demand increase 比 supply increase 大，价格上升；若 supply increase 更大，价格下降；幅度相同则价格可能不变。图中可以画两种可能的新交点，或在文字中清楚说明条件。

知识点 03/06 | 需求和供给同时移动中的主体、变量与判断对象 | Agents, variables and object of analysis

当 D 与 S 同时移动，equilibrium price 和 quantity 分别受到两个方向影响。先隔离每一次 shift，再把对同一结果变量的方向合并；同向可确定，反向则取决于 relative magnitude。

知识点 04/06 | 需求和供给同时移动的因果机制 | How the mechanism works

healthy diet campaign → tomato demand right → P ↑ Q ↑；good harvest → tomato supply right → P ↓ Q ↑；合并后 Q ↑，P depends on relative shifts。

知识点 05/06 | 需求和供给同时移动的图表与数据读法 | Diagram and data interpretation

先只移动 demand 并记录价格、数量方向，再恢复原图只移动 supply，最后把两个结果合并。若两个变化对某一变量方向相反，就只能写 depends on the relative size of the shifts。

知识点 06/06 | 需求和供给同时移动的关键英文表述 | Exam-ready explanation in English

Do not add or average separate equilibrium prices. The final equilibrium is the intersection of the two new curves. State the certain variable first and then explain why the other is indeterminate.

逐题精讲与评价拓展 | Exam workshop (共 3 项)

精讲 01/03 | 选择题: 区分最接近的概念 | Multiple-choice application

Question. Demand falls while supply rises. Which outcome is certain?

正确答案 | Correct answer. Equilibrium price falls. Both shifts put downward pressure on price, while their effects on quantity oppose each other.

精讲 02/03 | 结构题: 把情境发展成完整分析 | Structured application

Question. Analyse the effect on the market for electric cars of a successful advertising campaign and a fall in battery costs.

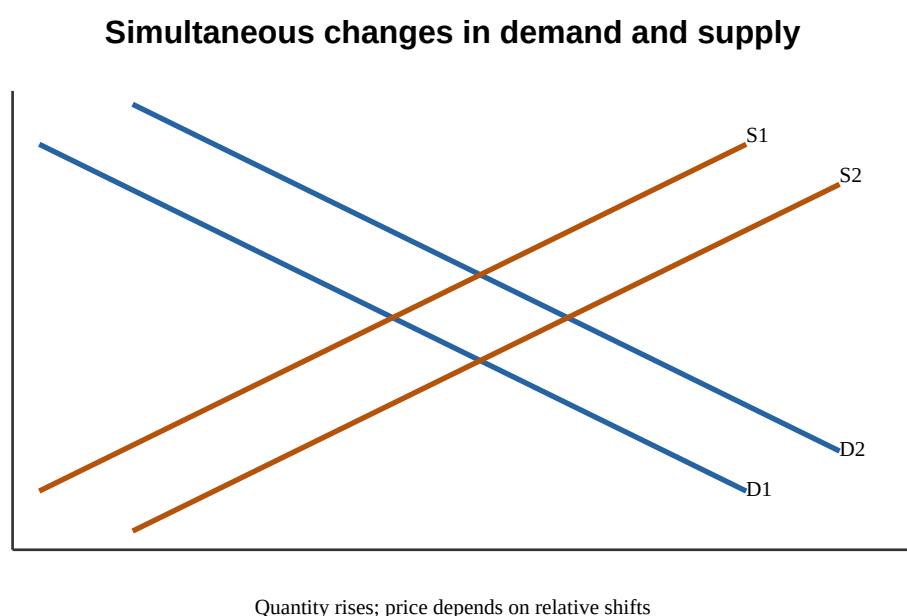
参考答案 | Model answer. Advertising shifts demand right. Lower battery costs shift supply right. Both changes increase equilibrium quantity. Demand growth raises price while supply growth lowers it, so the final price depends on the relative size of the shifts.

精讲 03/03 | 图表或计算题: 把方法落实到步骤 | Diagram or calculation application

Question. Demand and supply both increase. Which equilibrium result is certain, and which is uncertain? Use a diagram to explain. [5]

正确答案 | Correct answer. Shift D and S to the right. Equilibrium quantity certainly rises because both changes raise quantity. Equilibrium price is uncertain because the demand increase raises price while the supply increase lowers it; the final direction depends on the relative size of the shifts.

图表推演 | Diagram analysis



考试图表必须标出坐标轴、曲线名称、新旧均衡和变化方向。文字解释需说明原因、曲线移动以及价格或数量结果。

第 12 课 | 2.7 PED、PES 与销售收入 | Elasticity and revenue

所属大章: 资源配置 | The allocation of resources

知识点精讲 | Core knowledge

PED = percentage change in quantity demanded $\div$ percentage change in price。绝对值大于 1 为 elastic, 小于 1 为 inelastic, 等于 1 为 unit elastic。对缺乏弹性的需求, 提高价格会使 total revenue 上升; 对富有弹性的需求, 提高价格会使 total revenue 下降。PES 取决于闲置产能、库存、生产周期和要素流动性。

English explanation. Price elasticity measures responsiveness, not the direction of a curve shift. Always show percentage changes and interpret the value in context. A firm facing elastic demand may raise revenue by lowering price.

深度展开与真题化应用 | In-depth development (共 9 个知识点)

知识点 01/09 | PED、PES 与销售收入: 弹性衡量“反应有多大” | Elasticity measures the size of a response

PED 使用百分比而非绝对变化, 使不同货币和数量单位可以比较。通常取绝对值分类: $|PED| > 1$ elastic, $|PED| < 1$ inelastic。必需品、替代品少、支出占比低以及短期通常使需求较缺乏弹性; 奢侈品、替代品多和较长时间使需求较富弹性。PES 则取决于 spare capacity、stocks、production period 和 factor mobility。

知识点 02/09 | PED、PES 与销售收入: PED 与 total revenue | Connecting PED with total revenue

$TR = P \times Q$ 。若需求缺乏弹性, 价格提高的百分比大于数量减少的百分比, TR 上升; 若需求富有弹性, 数量减少比例更大, TR 下降。计算时例如 P 由 10 增至 12 (20%), Q 由 100 降至 70 (30%), $PED = 30/20 = 1.5$, 属 elastic, TR 从 1000 降至 840。

知识点 03/09 | PED、PES 与销售收入中的主体、变量与判断对象 | Agents, variables and object of analysis

PED 衡量 consumers 的 quantity demanded 对 price 的反应; PES 衡量 producers 的 quantity supplied 对 price 的反应。两者都使用 percentage change, 但决定因素与应用对象不同。

知识点 04/09 | PED、PES 与销售收入的因果机制 | How the mechanism works

many close substitutes $\rightarrow$ consumers can switch easily after price rise $\rightarrow$ large percentage fall in $Q_d \rightarrow$ PED more elastic。spare capacity $\rightarrow$ firms expand quickly after price rise $\rightarrow$ larger percentage rise in $Q_s \rightarrow$ PES more elastic。

知识点 05/09 | PED、PES 与销售收入的数字或现实情境 | Numerical or real-world application

价格由 \$20 降至 \$18, 下降 10%; Q_d 由 200 增至 250, 增加 25%; $|PED| = 25/10 = 2.5$, elastic。TR 由 4000 增至 4500, 符合降价提高富弹性需求下收入。

知识点 06/09 | PED、PES 与销售收入的图表与数据读法 | Diagram and data interpretation

弹性不是单看曲线视觉斜率。先写百分比变化公式, 再结合 substitutes、time period、necessity 和 capacity 解释数值。收入题还要把 price、quantity 与 $total\ revenue = P \times Q$ 连接。

知识点 07/09 | PED、PES 与销售收入的关键英文表述 | Exam-ready explanation in English

Elasticity measures responsiveness, not slope alone. The same straight demand curve has different elasticity at different points. Determinants include substitutes, necessity, budget share and time; for supply, capacity, stock and production time are central.

知识点 08/09 | PED、PES 与销售收入最容易混淆的口径 | Common misconception

弹性没有货币单位。不要把“2”解释为数量增加2个；应写 quantity changes by about 2% for a 1% price change。

知识点 09/09 | PED、PES 与销售收入结论成立的条件 | When the conclusion may change

企业收入结论要求其他条件不变。竞争者反应、品牌差异和成本变化可能让实际 revenue 与简单 PED 预测不同。

逐题精讲与评价拓展 | Exam workshop (共 3 项)

精讲 01/03 | 选择题: 区分最接近的概念 | Multiple-choice application

Question. Price rises by 8% and quantity demanded falls by 4%. What is PED and what happens to total revenue?

正确答案 | Correct answer. PED is 0.5 in absolute value, so demand is inelastic and total revenue is likely to rise.

精讲 02/03 | 结构题: 把情境发展成完整分析 | Structured application

Question. Analyse why demand for a life-saving medicine may be price inelastic and how this affects a tax.

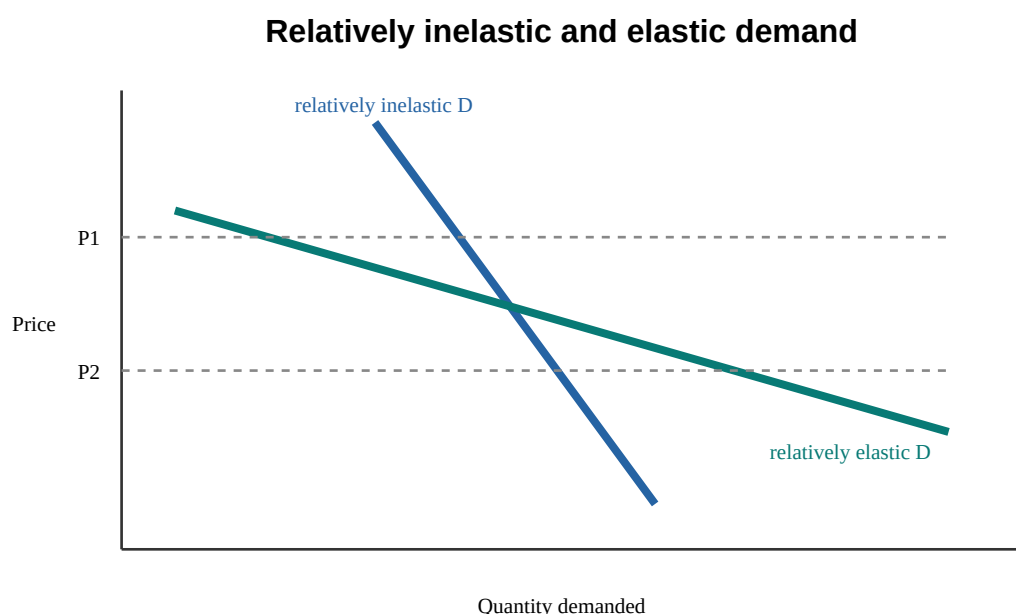
参考答案 | Model answer. The medicine is a necessity and may have few substitutes, so patients reduce purchases by a smaller percentage than the price rise. A tax can therefore raise substantial revenue, but consumption and the related external cost may fall only slightly. Low-income patients may bear a large burden.

精讲 03/03 | 图表或计算题: 把方法落实到步骤 | Diagram or calculation application

Question. Price falls by 10% and quantity demanded rises by 25%. Calculate PED and explain what happens to total revenue. [4]

正确答案 | Correct answer. $PED = 25\% / -10\% = -2.5$, or 2.5 in absolute value. Demand is price-elastic. Quantity rises proportionately more than price falls, so total revenue rises, other things equal.

图表推演 | Diagram analysis



考试图表必须标出坐标轴、曲线名称、新旧均衡和变化方向。文字解释需说明原因、曲线移动以及价格或数量结果。

第 13 课 | 2.8 市场失灵与政府干预 | Market failure and intervention

所属大章: 资源配置 | The allocation of resources

知识点精讲 | Core knowledge

市场失灵指自由市场没有把资源配置到社会最优位置, 原因包括 external costs and benefits、public goods、information failure 和 market power。政策包括 indirect taxes、subsidies、regulation、information provision、price controls 和 direct provision。

English explanation. Market failure occurs when the price mechanism does not allocate resources efficiently. A tax may reduce consumption of a demerit good, but effectiveness depends on PED, enforcement costs and unintended effects.

深度展开与真题化应用 | In-depth development (共 8 个知识点)

知识点 01/08 | 市场失灵与政府干预: 从私人决定到社会结果 | From private decisions to social outcomes

市场失灵发生在 free market allocation 不能最大化社会福利。负外部性中, 生产者或消费者只考虑 private cost, 却把污染、噪声或健康损害转嫁给 third parties, 因此 $\text{social cost} > \text{private cost}$, 市场数量可能过多。正外部性中 $\text{social benefit} > \text{private benefit}$, 教育或疫苗可能被少消费。

知识点 02/08 | 市场失灵与政府干预: 不同失灵需要不同政策 | Matching intervention to the source of market failure

Public goods 具有 non-excludability 和 non-rivalry, free-rider problem 使私人企业难收费, 政府可能直接提供。Information failure 使消费者低估有益品或高估有害品。Indirect tax、subsidy、regulation、information campaign、maximum/minimum price 和 direct provision 各有成本、时滞与副作用。

知识点 03/08 | 市场失灵与政府干预中的主体、变量与判断对象 | Agents, variables and object of analysis

Private decision-makers 依据 private costs and benefits; 社会评价还包括 external costs and benefits。若市场价格没有反映第三方影响, market quantity 可能偏离 socially efficient quantity。

知识点 04/08 | 市场失灵与政府干预的因果机制 | How the mechanism works

factory pollution $\rightarrow$ external cost imposed on residents $\rightarrow$ private cost below social cost $\rightarrow$ market price too low/output too high $\rightarrow$ indirect tax raises private cost $\rightarrow$ supply left $\rightarrow$ output and pollution may fall。

知识点 05/08 | 市场失灵与政府干预的数字或现实情境 | Numerical or real-world application

每单位产品 private cost=\$6, 污染 external cost=\$3, 则 social cost=\$9。若市场只按 \$6 作决定, 会忽略 \$3 的第三方损害; 理论上每单位 \$3 的税可使决策者面对更完整成本。

知识点 06/08 | 市场失灵与政府干预的关键英文表述 | Exam-ready explanation in English

An externality affects a third party not directly involved in the transaction. Public goods are non-excludable and non-rival, creating a free-rider problem. Information failure occurs when choices are based on inaccurate or incomplete information.

知识点 07/08 | 市场失灵与政府干预最容易混淆的口径 | Common misconception

Public good 不等于“政府提供的好东西”。必须同时具有 non-excludability 与 non-rivalry。

知识点 08/08 | 市场失灵与政府干预结论成立的条件 | When the conclusion may change

政府干预也有 information、administrative cost、unintended consequence 和 political pressure，评价时比较 market failure 与 government failure。

逐题精讲与评价拓展 | Exam workshop (共 2 项)

精讲 01/02 | 选择题: 区分最接近的概念 | Multiple-choice application

Question. Which is the clearest example of an external cost: a firm pays wages, a smoker harms nearby non-smokers, a consumer buys food, or a bank charges interest?

正确答案 | Correct answer. Harm to nearby non-smokers, because the cost falls on third parties outside the transaction.

精讲 02/02 | 结构题: 把情境发展成完整分析 | Structured application

Question. Discuss whether a tax is the best way to reduce consumption of sugary drinks.

参考答案 | Model answer. A tax raises firms' costs and price, reducing quantity demanded and health external costs while raising revenue. However, demand may be inelastic, the tax may be regressive and substitutes may also be unhealthy. Information campaigns or regulation may complement the tax. Effectiveness depends on tax size and alternatives.

Unit 3 | 微观经济决策者

Microeconomic decision makers

单元导读 | Unit introduction

家庭、劳动者、银行与企业都在约束下作决定。本单元连接 money、banking、household spending and saving、wages、trade unions、production、costs、revenue、profit、economies of scale 与 market structure。公式不是孤立计算：每个数值都要解释它怎样改变企业或家庭的激励。

学习主线 | Learning route

做企业题时先写清 output、cost、revenue 与 profit 的关系；做劳动市场题时区分 wage rate、earnings、productivity 与 employment。评价企业扩张或工会影响时，必须同时看短期成本与长期生产率、不同利益相关者和市场竞争程度。

English overview

Households supply labour, consume and save. Banks connect savers and borrowers. Firms combine factors of production and compare revenue with cost. The same change can create gains for one group and costs for another, so explanation and evaluation must identify the affected decision-maker.

本单元课时 | Lessons in this unit

第 14 课 | 2.1 货币与银行 (Money and banking)；第 15 课 | 2.2 家庭、劳动者与工会 (Households, workers and trade unions)；第 16 课 | 2.3 企业的生产、成本与收益；第 17 课 | 2.4 规模经济与市场结构；第 18 课 | 2.5 从成本表到企业决策

第 14 课 | 2.1 货币与银行 (Money and banking)

所属大章: 微观经济决策者 | Microeconomic decision makers

知识点精讲 | Core knowledge

货币的四项功能是 medium of exchange (交换媒介)、measure of value (价值尺度)、store of value (价值储藏) 与 standard of deferred payment (延期支付标准)。商业银行 (commercial bank) 接受存款、提供支付服务并放贷; 中央银行 (central bank) 通常发行货币、监管银行并影响利率。通胀削弱货币的购买力, 因此 “钱可以储藏价值” 不等于 “价值永远不变”。

深度展开与真题化应用 | In-depth development (共 7 个知识点)

知识点 01/07 | 货币与银行: 货币功能与良好货币特征 | Functions and characteristics of money

货币作为 medium of exchange 消除 barter 的 double coincidence of wants; 作为 measure of value 让商品可比较; 作为 store of value 把购买力转移到未来; 作为 standard of deferred payment 支持借贷。良好货币应 durable、portable、divisible、recognisable、scarce and stable in value。高通胀会削弱价值储藏功能。

知识点 02/07 | 货币与银行: 商业银行与中央银行 | Commercial-bank and central-bank roles

商业银行接受 deposits、提供 loans、支付和外汇服务, 并通过利差盈利。中央银行通常发行货币、监管银行、作为政府银行和最后贷款人, 并实施 monetary policy。利率上升提高借款成本, 可能减少消费和投资; 同时提高储蓄回报。结果取决于信心、债务类型和利率变化幅度。

知识点 03/07 | 货币与银行中的主体、变量与判断对象 | Agents, variables and object of analysis

Households 持有货币和存款, commercial banks 接受存款并提供贷款与支付, central bank 管理货币体系与利率。必须区分 money functions、bank functions 与 interest-rate effects。

知识点 04/07 | 货币与银行的因果机制 | How the mechanism works

central bank raises interest rate $\rightarrow$ borrowing becomes dearer/saving more attractive $\rightarrow$ household consumption and firm investment may fall $\rightarrow$ AD falls $\rightarrow$ inflationary pressure may ease。

知识点 05/07 | 货币与银行的数字或现实情境 | Numerical or real-world application

存款 \$1 000、年利率3%, 一年利息约 \$30。若通胀5%, 实际购买力仍可能下降约2%, 说明 nominal return 与 real return 不同。

知识点 06/07 | 货币与银行的关键英文表述 | Exam-ready explanation in English

Money performs four functions: medium of exchange, measure of value, store of value and standard of deferred payment. Commercial banks accept deposits and make loans. The central bank conducts monetary policy, supervises the banking system and may act as lender of last resort.

知识点 07/07 | 货币与银行最容易混淆的口径 | Common misconception

中央银行通常不为普通家庭办理日常存款; 商业银行也不负责制定全国政策利率。

逐题精讲与评价拓展 | Exam workshop (共 2 项)

精讲 01/02 | 选择题: 区分最接近的概念 | Multiple-choice application

Question. Which function of money is weakened most directly by rapid inflation?

正确答案 | Correct answer. Store of value, because money loses purchasing power quickly over time.

精讲 02/02 | 结构题: 把情境发展成完整分析 | Structured application

Question. Analyse how a rise in interest rates may affect households and firms.

参考答案 | Model answer. Loan and mortgage repayments may rise, reducing disposable income and consumption. Saving becomes more attractive. Firms face a higher cost of borrowing, so fewer investment projects are profitable. Demand and employment may fall, although savers receive more interest.

第 15 课 | 2.2 家庭、劳动者与工会 (Households, workers and trade unions)

所属大章: 微观经济决策者 | Microeconomic decision makers

知识点精讲 | Core knowledge

家庭提供 labour、land、capital, 获得收入后在消费与储蓄之间选择。劳动需求是 derived demand (派生需求) : 企业需要工人, 是因为消费者需要工人生产的商品。工资、技能、资格、工作条件、歧视和工会谈判力量都会影响工资差异。工会可用集体谈判提高工资和安全, 也可能在工资高于均衡时增加失业; 要写“可能”, 不能绝对化。

深度展开与真题化应用 | In-depth development (共 8 个知识点)

知识点 01/08 | 家庭、劳动者与工会: 家庭如何分配收入 | How households allocate income

家庭收入可用于 consumption、saving 和 direct taxes。收入越高, 通常能储蓄的绝对额越大, 但比例还受利率、信心、年龄、财富和未来预期影响。消费者选择也受 price、income、advertising、quality 和 tastes 影响。借贷让当前消费增加, 但未来必须偿还本息。

知识点 02/08 | 家庭、劳动者与工会: 工资差异与劳动市场 | Wage differences and the labour market

工资由劳动需求与供给共同影响。需要高技能、培训期长、风险大或工作条件差的职业可能工资较高; 工会、最低工资、歧视和公共部门政策也会改变工资。Trade union 通过 collective bargaining 改善工资和条件, 但工资高于生产率增长可能增加成本、价格或失业风险。

知识点 03/08 | 家庭、劳动者与工会中的主体、变量与判断对象 | Agents, variables and object of analysis

Households 同时是 consumers、savers 与 labour suppliers; workers 关注 wage、conditions 和 job security; trade unions 通过 collective bargaining 改变谈判力量。

知识点 04/08 | 家庭、劳动者与工会的因果机制 | How the mechanism works

product demand rises → derived demand for labour rises → labour shortage strengthens worker bargaining power → wages may rise → firms' costs may increase, unless productivity also rises.

知识点 05/08 | 家庭、劳动者与工会的数字或现实情境 | Numerical or real-world application

月收入 \$2 000, direct tax \$300, 消费 \$1 400, 则 saving=\$300。利率上升不保证储蓄必然增加: income effect 与 substitution effect 可能方向不同。

知识点 06/08 | 家庭、劳动者与工会的关键英文表述 | Exam-ready explanation in English

Labour demand is derived because firms employ workers to produce goods and services. Workers compare wages with hours, conditions, security, promotion and location. Trade unions use collective bargaining and may also provide legal or training services.

知识点 07/08 | 家庭、劳动者与工会最容易混淆的口径 | Common misconception

Labour supply increase 是更多工人愿意提供劳动; employment increase 是实际受雇人数增加, 两者不是同义词。

知识点 08/08 | 家庭、劳动者与工会结论成立的条件 | When the conclusion may change

工资高低还应与 real wage、工作时长、生活成本、福利和就业稳定性一起判断。

逐题精讲与评价拓展 | Exam workshop (共 2 项)

精讲 01/02 | 选择题: 区分最接近的概念 | Multiple-choice application

Question. Which change is most likely to increase demand for chefs: lower demand for restaurant meals, higher chef productivity, more people entering chef training, or lower food prices with no sales change?

正确答案 | Correct answer. Higher chef productivity, because each chef contributes more output and revenue to the firm.

精讲 02/02 | 结构题: 把情境发展成完整分析 | Structured application

Question. Discuss whether trade-union action always benefits workers.

参考答案 | Model answer. Collective bargaining may raise wages, improve safety and reduce exploitation. Higher pay can motivate workers and reduce turnover. However, large wage increases can raise costs, encourage automation or reduce employment. Non-members and consumers may lose. The result depends on productivity and labour-demand elasticity.

第 16 课 | 2.3 企业的生产、成本与收益

所属大章：微观经济决策者 | Microeconomic decision makers

知识点精讲 | Core knowledge

Fixed costs (租金、厂房) 不随短期产量变化; variable costs (原料、按件工资) 会变化。Total cost = fixed cost + variable cost。Total revenue = price $\times$ quantity; profit = total revenue - total cost。企业不只追求利润, 也可能追求 sales revenue、market share、survival 或环境声誉; 题目给了目标才应据此分析。

深度展开与真题化应用 | In-depth development (共 7 个知识点)

知识点 01/07 | 企业的生产、成本与收益: 生产率、分工与生产阶段 | Productivity, specialisation and stages of production

生产率是单位投入产出, 例如 output per worker, 不等于总产量。Division of labour 把生产拆成任务, 使工人 specialised, 减少转换时间并便于使用机器; 但重复工作可能降低 motivation, 停工也会影响整条生产线。Primary、secondary、tertiary sectors 分别涉及资源提取、制造建筑和服务。

知识点 02/07 | 企业的生产、成本与收益: 成本收益必须按公式连接 | Linking costs, revenue and profit with formulas

FC 不随 output 改变, VC 随 output 改变; $TC=FC+VC$, $AC=TC/Q$ 。TR= $P \times Q$, profit=TR-TC。若产量由 100 增至 200, TC 从 1000 增至 1600, AC 从 10 降至 8, 说明平均成本下降, 即使总成本上升。利润变化必须同时看 price、quantity 和 cost。

知识点 03/07 | 企业的生产、成本与收益中的主体、变量与判断对象 | Agents, variables and object of analysis

Firms 把 factors of production 转化为 output。Productivity 连接投入与产出; cost、revenue 与 profit 连接生产决定与财务结果; 短期至少一种要素固定, 长期所有要素可变。

知识点 04/07 | 企业的生产、成本与收益的因果机制 | How the mechanism works

division of labour $\rightarrow$ repeated practice and less switching time $\rightarrow$ output per worker rises $\rightarrow$ unit cost may fall $\rightarrow$ price may fall or profit rise $\rightarrow$ sales/investment may increase。

知识点 05/07 | 企业的生产、成本与收益的数字或现实情境 | Numerical or real-world application

$Q=100$, $FC=\$400$, $VC=\$600$, 所以 $TC=\$1\,000$ 、 $AC=\$10$ 。若 price= $\$14$, TR= $\$1\,400$ 、profit= $\$400$ 。Q 增加后总成本可上升而平均成本下降。

知识点 06/07 | 企业的生产、成本与收益的关键英文表述 | Exam-ready explanation in English

Production is the transformation of inputs into output. Division of labour can raise productivity through specialisation, but repetitive work may reduce motivation. Profit is a residual: total revenue minus total cost, not the same as sales revenue.

知识点 07/07 | 企业的生产、成本与收益最容易混淆的口径 | Common misconception

FC 不随 output 改变, 但 per-unit fixed cost 会变; VC 随 output 变, 不代表每单位 VC 必须恒定。

逐题精讲与评价拓展 | Exam workshop (共 2 项)

精讲 01/02 | 选择题: 区分最接近的概念 | Multiple-choice application

Question. A firm's output doubles while total cost rises by 50%. What happens to average cost?

正确答案 | Correct answer. Average cost falls, because cost rises less proportionately than output.

精讲 02/02 | 结构题: 把情境发展成完整分析 | Structured application

Question. Analyse two ways improved labour productivity may affect a firm.

参考答案 | Model answer. More output per worker can reduce unit labour cost, allowing a lower price and greater competitiveness. Alternatively, the firm may keep price unchanged and earn more profit, financing investment. The effect depends on wages, product quality and demand.

第 17 课 | 2.4 规模经济与市场结构

所属大章: 微观经济决策者 | Microeconomic decision makers

知识点精讲 | Core knowledge

长期扩张时, 平均成本下降叫 economies of

scale: 可以来自批量采购、专业化、管理、技术和融资。规模太大可能产生 communication problems 和 coordination problems, 即 diseconomies of scale。市场结构从竞争强到弱大致为 perfect

competition、monopolistic competition、oligopoly、monopoly; 答题不要只背定义, 应把进入壁垒、产品差异、定价能力和消费者选择联系起来。

深度展开与真题化应用 | In-depth development (共 8 个知识点)

知识点 01/08 | 规模经济与市场结构: 企业变大为何可能先降成本 | Why expansion may initially lower average cost

Internal economies of scale 包括 purchasing、technical、financial、managerial、marketing 和 risk-bearing economies。大型企业批量采购可议价, 专用机器产量大时才划算, 借款风险较低, 专业经理提高决策质量, 广告成本可分摊到更多产品。它们影响 long-run average cost, 不等同于短期固定成本分摊。

知识点 02/08 | 规模经济与市场结构: 规模不经济与竞争程度 | Diseconomies of scale and market competition

企业过大可能出现 communication failure、协调困难、管理层级增加和员工疏离, 使 average cost 上升。竞争市场中 firms 多、进入较易, 企业面临降价和创新压力; monopoly 有规模优势和研发资金, 也可能提高价格、限制选择或降低效率。市场份额高不自动证明滥用市场力量。

知识点 03/08 | 规模经济与市场结构中的主体、变量与判断对象 | Agents, variables and object of analysis

Internal economies 来自单个企业扩大, external economies 来自整个行业或地区扩大。市场结构则看 firms number、entry barriers、product differentiation 与 market power, 不能只用企业规模判断。

知识点 04/08 | 规模经济与市场结构的因果机制 | How the mechanism works

bulk purchasing → stronger bargaining power → lower input price per unit → average cost falls → firm can lower price → market share may rise → further scale economies may develop。

知识点 05/08 | 规模经济与市场结构的数字或现实情境 | Numerical or real-world application

小企业年广告费 \$20 000、销量 10 000, 广告成本每件 \$2; 大企业广告费 \$100 000、销量 100 000, 每件 \$1。这是 marketing economy, 但总广告费更高。

知识点 06/08 | 规模经济与市场结构的关键英文表述 | Exam-ready explanation in English

Internal economies arise from the firm's own expansion; external economies arise when the industry or region expands. Diseconomies often result from communication, coordination and motivation problems in very large organisations.

知识点 07/08 | 规模经济与市场结构最容易混淆的口径 | Common misconception

规模经济是 average cost 下降, 不是 total cost 下降; market share 大也不等于只有一家企业。

知识点 08/08 | 规模经济与市场结构结论成立的条件 | When the conclusion may change

Natural monopoly

中重复基础设施成本高，受监管单一企业可能比许多小企业有效；其他市场则竞争更可能保护消费者。

逐题精讲与评价拓展 | Exam workshop (共 2 项)

精讲 01/02 | 选择题: 区分最接近的概念 | Multiple-choice application

Question. Which is an internal economy of scale: lower national interest rates, a skilled regional labour pool, bulk-buying discounts, or a new public road?

正确答案 | Correct answer. Bulk-buying discounts, because they result directly from the firm's larger purchasing scale.

精讲 02/02 | 结构题: 把情境发展成完整分析 | Structured application

Question. Discuss whether consumers benefit from a monopoly.

参考答案 | Model answer. A monopoly may exploit scale economies, fund research and provide a standardised network. It may pass lower costs to consumers. Conversely, weak competition may permit higher prices, less choice and lower quality. Outcomes depend on regulation, contestability and the source of monopoly power.

第 18 课 | 2.5 从成本表到企业决策

所属大章：微观经济决策者 | Microeconomic decision makers

知识点精讲 | Core knowledge

把题中的每一列先翻译：TC 是总成本； $AC = TC/Q$ ； $TR = P \times Q$ ； $profit = TR - TC$ 。若固定成本不变，产量上升时 AC 可能下降，因为固定成本被更多单位分摊；这不代表每一项成本下降。

例子：厂房租金每月 10 000，生产 1 000 件时每件固定成本 10；生产 2 000 件时每件固定成本 5。若增加产量需要加班和更昂贵的原料，variable cost 仍可能上升。因此“规模扩大一定更便宜”错误。

写工会题时分两条线：工资提高会提升会员收入；企业劳动力成本提高可能减少雇佣、提高价格或投资自动化。结论取决于 labour demand elasticity、生产率和企业利润率。

深度展开与真题化应用 | In-depth development (共 5 个知识点)

知识点 01/05 | 从成本表到企业决策：读表顺序决定是否出错 | A reliable order for reading a cost table

先识别 output，再找固定成本是否在各产量相同；计算 $VC = TC - FC$ ；再计算 $AC = TC / \text{output}$ 、 $TR = \text{price} \times \text{quantity}$ 和 $profit = TR - TC$ 。零产量时 TC 通常等于 FC，但 AC 无法用 0 作除数。不要因为 AC 下降就写 TC 下降，也不要因为 revenue 当 profit。

知识点 02/05 | 从成本表到企业决策：数据变化要解释原因 | Explaining why cost and revenue figures change

AC 下降可能来自固定成本分摊、specialisation 或更高 machinery utilisation；随后上升可能来自 overtime、拥堵和协调问题。利润最大并不一定发生在 output 最大处，因为额外产量也带来额外成本。IGCSE 通常不要求边际分析，但必须比较收益与成本。

知识点 03/05 | 从成本表到企业决策中的主体、变量与判断对象 | Agents, variables and object of analysis

FC 不随 output 变化，VC 随 output 变化， $TC = FC + VC$ ； $AC = TC / Q$ ， $MC = \Delta TC / \Delta Q$ ； $TR = P \times Q$ ， $profit = TR - TC$ 。每个公式回答不同问题。

知识点 04/05 | 从成本表到企业决策的数字或现实情境 | Numerical or real-world application

$Q=0$ 时 $FC=240$ 、 $TC=240$ 。 $Q=40$ 时 $TC=600$ ，所以 $VC=360$ 、 $AC=15$ 。 $\text{price}=18$ 时 $TR=720$ 、 $profit=120$ 。若把 revenue 720 当 profit，会忽略全部成本。

知识点 05/05 | 从成本表到企业决策最容易混淆的口径 | Common misconception

零产量时 AC 不可计算，因为不能除以零；profit 可能为负，称 loss。

逐题精讲与评价拓展 | Exam workshop (共 3 项)

精讲 01/03 | 选择题：区分最接近的概念 | Multiple-choice application

Question. At output 50, total cost is \$900 and variable cost is \$650. What is fixed cost?

正确答案 | Correct answer. \$250. Fixed cost equals total cost minus variable cost.

精讲 02/03 | 结构题：把情境发展成完整分析 | Structured application

Question. Analyse why a firm may continue producing even when total profit falls temporarily.

参考答案 | Model answer. It may expect demand to recover, wish to retain skilled workers or avoid high closure and restart costs. If revenue covers variable cost and contributes to fixed cost, short-run production can reduce losses. In the long run, persistent revenue below total cost is unsustainable.

精讲 03/03 | 图表或计算题: 把方法落实到步骤 | Diagram or calculation application

Question. Output rises from 40 to 50 units and total cost rises from \$600 to \$720. Calculate marginal cost per extra unit and explain what it measures. [3]

正确答案 | Correct answer. The extra cost is $\$720 - \$600 = \$120$ for 10 additional units, so marginal cost per extra unit is \$12. It measures the addition to total cost caused by producing one more unit over this interval.

Unit 4 | 政府与宏观经济

Government and the macroeconomy

单元导读 | Unit introduction

本单元从 national aims and indicators 开始，解释 economic growth、employment、price stability、redistribution 与 balance of payments。随后学习 fiscal、monetary 与 supply-side policies。分析政策时不能从“降息”直接跳到“经济增长”，而要写出借款成本、消费或投资、总需求、产出与价格的传导。

学习主线 | Learning route

宏观题先分清名义值与实际值、总量与人均量、单个价格变化与持续通胀。政策评价从 time lag、confidence、spare capacity、elasticity、government finance 与 side effects 中选择真正相关的条件。

English overview

Macroeconomic indicators describe different parts of an economy. Real GDP measures output after removing price changes; unemployment and inflation require separate explanations. Policy answers must develop the transmission mechanism and then judge whether it is likely to be strong, weak or delayed.

本单元课时 | Lessons in this unit

第 19 课 | 2.1 宏观目标与指标；第 20 课 | 2.2 经济增长、失业与通胀；第 21 课 | 2.3 政策工具 (Policy tools) ；第 22 课 | 2.4 国际收支 (Balance of payments) ；第 23 课 | 2.5 政策传导链的写法

第 19 课 | 2.1 宏观目标与指标

所属大章：政府与宏观经济 | Government and the macroeconomy

知识点精讲 | Core knowledge

常见目标是经济增长、低失业、低且稳定通胀、国际收支平衡与更公平的收入分配。它们会冲突：刺激总需求或许降低周期性失业，但也可能提高通胀和进口，恶化经常账户。

深度展开与真题化应用 | In-depth development (共 7 个知识点)

知识点 01/07 | 宏观目标与指标：五类目标与测量口径 | Five macroeconomic objectives and their indicators

主要目标包括 economic growth、low unemployment、price stability、redistribution of income 和 balance of payments stability。Growth 用 real GDP 的变化衡量；通胀用 consumer price index 的持续增长；失业率=unemployed/labour force $\times 100$ ；收入分配可结合 Lorenz curve；国际交易重点看 current account。

知识点 02/07 | 宏观目标与指标：目标之间既有协同也有冲突 | Synergies and conflicts between policy objectives

衰退时提高 AD 可能同时增加 output 和 employment；接近满负荷时继续刺激 AD 更可能推高 inflation。快速增长可能增加 imports、恶化 current account，也可能通过生产率和出口改善它。重新分配收入可能提高低收入者消费，却可能影响工作、储蓄和投资激励。

知识点 03/07 | 宏观目标与指标中的主体、变量与判断对象 | Agents, variables and object of analysis

Government 常关注 growth、employment、price stability、current-account stability 与 income distribution。每个目标需要不同指标，指标改善也可能与另一目标冲突。

知识点 04/07 | 宏观目标与指标的因果机制 | How the mechanism works

AD rises when spare capacity exists $\rightarrow$ firms increase output $\rightarrow$ derived demand for labour rises $\rightarrow$ unemployment falls $\rightarrow$ household income and tax revenue rise；接近满产时则 inflation pressure 更大。

知识点 05/07 | 宏观目标与指标的数字或现实情境 | Numerical or real-world application

失业人数 0.6m、labour force 12m，unemployment rate=5%。若失业人数不变但更多人进入 labour force，失业率可能变化，因此分母也要检查。

知识点 06/07 | 宏观目标与指标的关键英文表述 | Exam-ready explanation in English

Macroeconomic objectives are related but not identical. Economic growth is measured using real output; price stability refers to low and predictable inflation; full employment does not mean zero unemployment because frictional unemployment normally exists.

知识点 07/07 | 宏观目标与指标结论成立的条件 | When the conclusion may change

比较国家时须使用相同年份、定义与单位，并注意 informal sector、underemployment 和地区差异。

逐题精讲与评价拓展 | Exam workshop (共 2 项)

精讲 01/02 | 选择题：区分最接近的概念 | Multiple-choice application

Question. Which policy outcome shows a conflict: lower unemployment accompanied by higher demand-pull inflation, or higher productivity with lower costs?

正确答案 | Correct answer. Lower unemployment with higher demand-pull inflation, because improvement in one objective is associated with deterioration in another.

精讲 02/02 | 结构题: 把情境发展成完整分析 | Structured application

Question. Discuss whether low unemployment is always beneficial to an economy.

参考答案 | Model answer. It raises output, income, tax revenue and may reduce benefit spending and poverty. However, severe labour shortages can increase wage costs and inflation, reduce competitiveness and make recruitment difficult. Some job movement is useful. The ideal rate depends on productivity and labour mobility.

第 20 课 | 2.2 经济增长、失业与通胀

所属大章: 政府与宏观经济 | Government and the macroeconomy

知识点精讲 | Core knowledge

Economic growth 是实际 GDP

增加; 它可提高收入与税收, 但若只依赖资源开采, 可能污染并加剧不平等。失业分为

frictional、structural、cyclical 和 seasonal。通胀是总体价格水平持续上升, 不是单一商品涨价; demand-pull inflation 来自总需求增长过快, cost-push inflation 来自工资、能源或进口成本上升。

深度展开与真题化应用 | In-depth development (共 8 个知识点)

知识点 01/08 | 经济增长、失业与通胀: 三个概念必须分开 | Distinguishing growth, unemployment and inflation

Economic growth 是 real GDP 增加; recession 通常是 real output 下降。Inflation

是总体价格水平持续上升, 不是一个商品涨价。Unemployment

指愿意且能够工作、正在寻找工作却没有工作的人。Frictional、structural、cyclical、seasonal unemployment 的原因和政策不同。

知识点 02/08 | 经济增长、失业与通胀: 原因与影响的双向链条 | Causes and consequences in both directions

Demand-pull inflation 来自 AD 增长过快; cost-push inflation 来自工资、能源或进口成本上升。通胀降低固定收入者购买力并增加不确定性, 但温和通胀也可能伴随增长。失业降低 household income、tax revenue 和 output, 却减少工资与价格压力。增长可来自资源增加、productivity、investment 和技术进步。

知识点 03/08 | 经济增长、失业与通胀中的主体、变量与判断对象 | Agents, variables and object of analysis

Growth 用 real GDP 变化衡量; inflation 用 general price level 的持续变化衡量; unemployment 只包括 willing and able to work 且正在找工作的人。Nominal、real、rate 与 level 必须区分。

知识点 04/08 | 经济增长、失业与通胀的因果机制 | How the mechanism works

oil price rises → firms' production costs rise → SRAS falls → price level rises and real output falls → cost-push inflation and cyclical unemployment may coexist.

知识点 05/08 | 经济增长、失业与通胀的数字或现实情境 | Numerical or real-world application

CPI 从120升到126, $\text{inflation} = (126 - 120) / 120 \times 100 = 5\%$ 。下一年 CPI

从126升到129.15, 通胀2.5%; 价格仍上涨, 只是上涨速度减慢。

知识点 06/08 | 经济增长、失业与通胀的关键英文表述 | Exam-ready explanation in English

Disinflation is a fall in the inflation rate while the price level continues to rise. Deflation is a sustained fall in the general price level. Structural unemployment reflects a mismatch of skills or location, while cyclical unemployment results from deficient aggregate demand.

知识点 07/08 | 经济增长、失业与通胀最容易混淆的口径 | Common misconception

Real GDP per head may fall even when real GDP rises if population grows faster.

知识点 08/08 | 经济增长、失业与通胀结论成立的条件 | When the conclusion may change

适度需求增长可能同时降失业并带来温和通胀；长期结果取决于 productive capacity、expectations 和政策可信度。

逐题精讲与评价拓展 | Exam workshop (共 2 项)

精讲 01/02 | 选择题: 区分最接近的概念 | Multiple-choice application

Question. The inflation rate falls from 8% to 3%. Which statement is correct?

正确答案 | Correct answer. The general price level is still rising, but more slowly. This is disinflation, not necessarily deflation.

精讲 02/02 | 结构题: 把情境发展成完整分析 | Structured application

Question. Analyse two costs of high inflation for firms.

参考答案 | Model answer. Uncertain future costs and prices make investment planning harder. If domestic inflation exceeds competitors', exports become less price competitive and imports more attractive. Firms may also face menu costs and pressure for higher wages.

第 21 课 | 2.3 政策工具 (Policy tools)

所属大章: 政府与宏观经济 | Government and the macroeconomy

知识点精讲 | Core knowledge

Fiscal policy: 政府增加支出或减税以提高 AD; 代价可能是预算赤字与债务。Monetary

policy: 降低利率使借贷更便宜, 往往提高消费和投资, 但在信心低迷时效果弱。Supply-side

policy: 培训、教育、基础设施与竞争政策提高生产能力, 见效较慢但更能改善长期增长。

深度展开与真题化应用 | In-depth development (共 8 个知识点)

知识点 01/08 | 政策工具: 三类政策作用点不同 | Where fiscal, monetary and supply-side policies act

Fiscal policy 改变 government spending 与 taxation; monetary policy 通过 interest rates、money/credit conditions 和 exchange rate 影响需求; supply-side policy 改善 education、infrastructure、competition、labour mobility 和 incentives, 提高 productive capacity。Expansionary 政策通常提高 AD, contractionary 政策降低 AD。

知识点 02/08 | 政策工具: 政策传导不是一句“促进经济” | Developing the policy transmission mechanism

降息 → borrowing cheaper/saving less attractive → C 与 I 可能增加 → AD rises → firms raise output → employment rises。所得税下降 → disposable income rises → consumption rises。培训 → skills/productivity rise → unit cost falls → AS/productive potential rises。每条链都要说明可能条件: confidence、time lag、size、elasticity、debt 和 import leakage。

知识点 03/08 | 政策工具中的主体、变量与判断对象 | Agents, variables and object of analysis

Fiscal policy 改变 government spending 与 taxation; monetary policy 改变 interest rates 或 credit; supply-side policy 改变资源数量、质量、流动性和生产率。工具必须匹配问题原因。

知识点 04/08 | 政策工具的因果机制 | How the mechanism works

income tax cut → disposable income rises → consumption may rise → AD rises → firms increase output and employment → inflation/imports may also rise。training → skills/productivity rise → productive capacity increases, but after a time lag。

知识点 05/08 | 政策工具的关键英文表述 | Exam-ready explanation in English

Fiscal policy changes government spending and taxation. Monetary policy changes interest rates or credit conditions. Supply-side policy aims to improve the quantity, quality or mobility of productive resources. Match the policy to the cause of the problem.

知识点 06/08 | 政策工具最容易混淆的口径 | Common misconception

Supply-side policy 不一定表示政府减少供给; 它旨在提高 economy's productive potential。

知识点 07/08 | 政策工具结论成立的条件 | When the conclusion may change

评价至少加入 size、timing、confidence、spare capacity、opportunity cost 与 distribution。

知识点 08/08 | 政策工具的综合应用 | Integrated application

Structural unemployment 来自技能或地区错配, retraining 比单纯降息更直接。降息主要提高 AD; 若空缺岗位要求的技能不同, 需求扩大也未必让失业者立即就业。

逐题精讲与评价拓展 | Exam workshop (共 2 项)

精讲 01/02 | 选择题: 区分最接近的概念 | Multiple-choice application

Question. Which policy is most directly designed to reduce structural unemployment: lower interest rates, retraining, higher government consumption, or a currency depreciation?

正确答案 | Correct answer. Retraining, because it reduces the mismatch between workers' skills and available jobs.

精讲 02/02 | 结构题: 把情境发展成完整分析 | Structured application

Question. Discuss whether higher government spending will increase economic growth.

参考答案 | Model answer. Spending raises AD and may increase output when spare capacity exists. Infrastructure and education can also increase long-run productivity. However, spending may cause inflation, imports, taxation or debt, and inefficient projects have opportunity cost. The result depends on spending type and economic conditions.

第 22 课 | 2.4 国际收支 (Balance of payments)

所属大章: 政府与宏观经济 | Government and the macroeconomy

知识点精讲 | Core knowledge

经常账户记录货物、服务、收入和转移支付。赤字代表流出大于流入，不必然坏：若进口的是资本设备，可能提升未来生产力；但长期靠短期借款融资会有风险。分析时区分“赤字的原因”和“赤字本身”。

深度展开与真题化应用 | In-depth development (共 7 个知识点)

知识点 01/07 | 国际收支: Current account 的组成与符号 | Current-account components and signs

Current account 包括 trade in goods、trade in services、primary income 和 secondary income。出口记录为 credit，进口为 debit。Surplus 表示流入大于流出，deficit 相反。它是流量，不是政府“欠款总额”，也不等同于财政预算赤字。

知识点 02/07 | 国际收支: 赤字原因和影响需结合背景 | Contextual causes and effects of a deficit

进口高可能来自高收入、强汇率或本国产品竞争力低；也可能是进口 capital goods，为未来出口能力打基础。赤字可能增加外债或使汇率承压，但居民可获得更多商品和机器。改善方法包括 depreciation、提高 productivity、贸易限制或降低 AD，各自有副作用。

知识点 03/07 | 国际收支中的主体、变量与判断对象 | Agents, variables and object of analysis

Current account 的主体是本国居民与外国之间的流量，项目包括 goods、services、primary income 和 secondary income。Credit 带来收入，debit 造成支付。

知识点 04/07 | 国际收支的因果机制 | How the mechanism works

domestic income rises → consumers buy more imports → import expenditure rises → current-account balance may deteriorate; 若增长来自 export productivity，结果可能相反。

知识点 05/07 | 国际收支的数字或现实情境 | Numerical or real-world application

Goods exports 80、imports 100; services exports 40、imports 25; net primary income -5、transfers +2。Current balance = $(80 - 100) + (40 - 25) - 5 + 2 = -8$ ，为 deficit 8。

知识点 06/07 | 国际收支的关键英文表述 | Exam-ready explanation in English

A current-account deficit means payments out exceed receipts in the recorded categories. It is not identical to government debt. A surplus may reflect strong exports, weak domestic demand or restricted imports, so its welfare effect is ambiguous.

知识点 07/07 | 国际收支最容易混淆的口径 | Common misconception

Trade balance 只含 goods (有些口径含 goods and services 时必须看题目); current account 范围更广。

逐题精讲与评价拓展 | Exam workshop (共 3 项)

精讲 01/03 | 选择题: 区分最接近的概念 | Multiple-choice application

Question. Which transaction is a credit item: a resident buys an imported car, a foreign tourist buys a hotel service domestically, or government pays foreign interest?

正确答案 | Correct answer. The foreign tourist's purchase, because it is an export of services and brings foreign currency receipts.

精讲 02/03 | 结构题: 把情境发展成完整分析 | Structured application

Question. Discuss whether a current-account deficit is harmful.

参考答案 | Model answer. It may weaken the currency, increase borrowing and indicate low competitiveness. Yet it may finance imported capital goods, reflect strong growth and provide consumers with choice. Harm depends on size, persistence, financing and whether imports improve future productivity.

精讲 03/03 | 图表或计算题: 把方法落实到步骤 | Diagram or calculation application

Question. Goods exports are 70, goods imports 95, services exports 40, services imports 25, net primary income -4 and net transfers +1. Calculate the current-account balance. [4]

正确答案 | Correct answer. $(70 - 95) + (40 - 25) - 4 + 1 = -13$. The economy has a current-account deficit of 13 units; the negative sign is a flow deficit, not government debt.

第 23 课 | 2.5 政策传导链的写法

所属大章：政府与宏观经济 | Government and the macroeconomy

知识点精讲 | Core knowledge

不要只写“降低利率促进经济”。拆开写：interest rate falls → monthly loan repayments fall / borrowing becomes cheaper → households consume more and firms invest more → AD rises → output and employment may rise.

然后评价：如果家庭担心失业，可能仍储蓄；若经济接近满产，效果更可能是通胀而不是实际产出。

财政政策同理：government spending on roads → firms receive orders → workers' incomes rise → further consumption occurs. 但政府融资可能增加债务；道路项目也有长期 supply-side benefit。把短期 AD 效果与长期 LRAS 效果分开，答案会更成熟。

深度展开与真题化应用 | In-depth development (共 6 个知识点)

知识点 01/06 | 政策传导链的写法：每个箭头都要有 because | Explaining every link in a causal chain

高分 analysis 不是列出变量，而是解释关系。例如 “interest rate ↓ → investment ↑ → AD ↑ → GDP ↑” 仍不够完整；应写借款成本下降提高预期净收益，使更多投资项目有利可图，capital spending 增加，AD 上升，企业面对更高销售而增加产量与就业。

知识点 02/06 | 政策传导链的写法：在链条末端加入限制条件 | Adding a relevant condition to the final outcome

若企业缺乏 confidence，即使利率下降也可能不借款；若经济接近 full employment，AD 增长更多转化为 inflation；若进口倾向高，本国 output 增幅较小。Supply-side policy 需要教育和基建完成后才见效，time lag 往往更长。

知识点 03/06 | 政策传导链的写法中的主体、变量与判断对象 | Agents, variables and object of analysis

政策工具先改变 price、income、cost 或 incentive，随后 households/firms 改变 spending、saving、investment 或 production，最终才影响 AD、output、employment 与 inflation。

知识点 04/06 | 政策传导链的写法的因果机制 | How the mechanism works

interest rate falls → monthly loan cost falls → households may borrow for durable goods → consumption rises → AD rises → firms receive more orders → output and employment rise, if spare capacity exists.

知识点 05/06 | 政策传导链的写法的数字或现实情境 | Numerical or real-world application

不要用没有题干依据的精确数字，但可以用方向检查。若税后收入由\$1 000升到\$1 100而家庭只消费新增\$60，剩余\$40 saving 是 leakage，AD 增量小于收入增量。

知识点 06/06 | 政策传导链的写法的关键英文表述 | Exam-ready explanation in English

A causal chain should not jump from policy to the final objective. Explain the transmission through incentives and behaviour. Separate a direct effect from secondary effects and avoid repeating the same link in different words.

逐题精讲与评价拓展 | Exam workshop (共 2 项)

精讲 01/02 | 选择题：区分最接近的概念 | Multiple-choice application

Question. Which is the missing link: lower interest rate → ____ → higher investment → higher AD?

正确答案 | Correct answer. Borrowing becomes cheaper or the expected net return on investment rises.

精讲 02/02 | 结构题: 把情境发展成完整分析 | Structured application

Question. Analyse how a depreciation may reduce unemployment.

参考答案 | Model answer. Exports become cheaper in foreign currency and imports become dearer domestically. If demand responds, export sales rise and spending switches to domestic products. Firms increase output, raising derived demand for labour and reducing cyclical unemployment.

Unit 5 | 经济发展

Economic development

单元导读 | Unit introduction

增长与发展不是同义词。real GDP 增加说明产出增长，但 living standards 还取决于人口、收入分配、健康、教育、环境与公共服务。本单元学习 indicators、poverty、population、inequality 与 development policies，并训练从数据描述走向因果解释。

学习主线 | Learning route

比较国家数据时先识别单位、年份与每人指标，再找支持关系和反例。讨论政策时要指出它缓解的是哪一种约束，例如 skills、health、infrastructure、finance 或 inequality，并说明覆盖范围、财政成本和见效时间。

English overview

Economic growth can create resources for development, but it does not guarantee better living standards. Per-head indicators, health and education data, poverty measures and distributional evidence must be considered together. Policy evaluation should identify the binding constraint and the groups that gain or lose.

本单元课时 | Lessons in this unit

第 24 课 | 2.1 增长不等于发展 (Growth versus development) ; 第 25 课 | 2.2 衡量生活水平 (Living standards) ; 第 26 课 | 2.3 贫困、人口与发展障碍; 第 27 课 | 2.4 发展策略的评价; 第 28 课 | 2.5 指标题的比较框架

第 24 课 | 2.1 增长不等于发展 (Growth versus development)

所属篇章: 经济发展 | Economic development

知识点精讲 | Core knowledge

增长是实际 GDP 上升; development (发展) 是生活质量和选择能力的更广泛改善。一个国家可因石油出口 GDP 增长, 却仍有贫困、教育不足、污染和地区差距。因此答案中把 “人均实际 GDP” 与健康、教育、住房、自由、安全、环境一起写。

深度展开与真题化应用 | In-depth development (共 7 个知识点)

知识点 01/07 | 增长不等于发展: 两个概念的范围不同 | Different meanings of growth and development

Growth 是 real GDP 增加, development 是生活质量的广泛改善, 包括 income、health、education、housing、environment 和 freedom。增长可增加税基和就业, 为发展提供资源; 但若收益集中在少数人、污染严重或人口增长更快, 平均居民未必受益。

知识点 02/07 | 增长不等于发展: 因果关系不是自动发生 | Why growth does not automatically create development

资源出口提高 GDP 后, 政府若把收入用于 clean water、schools 和 clinics, human capital 与 life expectancy 可提高; 若收入外流、腐败或只进入资本密集行业, poverty 可能仍高。Real GDP per head 比总 GDP 更接近平均物质生活, 但仍忽略 distribution 和非货币因素。

知识点 03/07 | 增长不等于发展中的主体、变量与判断对象 | Agents, variables and object of analysis

Growth 观察 real output; development 观察 income、health、education、opportunity、distribution 和 environment。GDP 是生产指标, 不是完整 welfare 指标。

知识点 04/07 | 增长不等于发展的因果机制 | How the mechanism works

export-led growth → employment/income/tax revenue rise → government may fund health and education → human capital and life expectancy improve → development; 若利润外流或污染严重, 链条可能中断。

知识点 05/07 | 增长不等于发展的数字或现实情境 | Numerical or real-world application

Real GDP 增6%、人口增4%, real GDP per head 大致增约2%, 不是6%。若最高10%获得大部分新增收入, median household 的生活水平可能几乎不变。

知识点 06/07 | 增长不等于发展的关键英文表述 | Exam-ready explanation in English

Growth expands the amount an economy produces. Development concerns what the output enables people to achieve. Per-head measures, distribution and non-material indicators are therefore essential when assessing welfare.

知识点 07/07 | 增长不等于发展结论成立的条件 | When the conclusion may change

结论最好区分 short run material gain 与 long-run sustainable development.

逐题精讲与评价拓展 | Exam workshop (共 2 项)

精讲 01/02 | 选择题: 区分最接近的概念 | Multiple-choice application

Question. Which most clearly indicates development rather than growth alone: higher real GDP, longer life expectancy, higher price level, or more currency in circulation?

正确答案 | Correct answer. Longer life expectancy, because it directly reflects a health and welfare outcome.

精讲 02/02 | 结构题: 把情境发展成完整分析 | Structured application

Question. Discuss whether rapid economic growth improves living standards.

参考答案 | Model answer. Growth may create jobs, raise incomes and tax revenue for services. It can finance infrastructure. However, population may grow faster, income may be unequal, work hours may rise and pollution or resource depletion may worsen. The result depends on inclusiveness and sustainability.

第 25 课 | 2.2 衡量生活水平 (Living standards)

所属大章: 经济发展 | Economic development

知识点精讲 | Core knowledge

GDP per head 适合粗略比较物质产出, 但忽略收入分配、非市场生产、工作时长与环境。HDI 结合寿命、教育和收入, 覆盖更广; 仍是平均数, 会掩盖性别、城乡与地区不平等。比较国家时还要注意汇率、购买力和人口结构。

深度展开与真题化应用 | In-depth development (共 6 个知识点)

知识点 01/06 | 衡量生活水平: 单一指标与综合指标 | Single indicators and composite measures

Real GDP per head=real GDP/population, 适合比较平均物质产出, 但跨国比较还要考虑 exchange rate、purchasing power、informal economy 和 data accuracy。HDI 结合 income、education 和 life expectancy, 范围更广, 却仍是平均值, 不能完整反映 inequality、security 或 environment。

知识点 02/06 | 衡量生活水平: 读数据要比较、联系并找例外 | Comparing data, identifying relationships and exceptions

一国人均收入高而 life expectancy

低, 可能说明不平等、医疗不足或污染; 另一国收入较低但教育健康良好, 可能源于公共服务效率。Correlation 不等于 causation, 还要考虑其他变量。比较时应使用同一年、相同单位和 real/per-head 数据。

知识点 03/06 | 衡量生活水平中的主体、变量与判断对象 | Agents, variables and object of analysis

Real GDP per head 调整 inflation 与 population; HDI 加入 education 与 life expectancy; 单项社会指标可补充 sanitation、nutrition 或 mortality。不同指标回答不同问题。

知识点 04/06 | 衡量生活水平的数字或现实情境 | Numerical or real-world application

A国 real GDP 500bn、人口50m, 人均10 000; B国300bn、人口20m, 人均15 000。A总产出更高, 但B平均物质产出更高。还不能据此判断分配和健康。

知识点 05/06 | 衡量生活水平的关键英文表述 | Exam-ready explanation in English

Real GDP per head is useful for average material output, while HDI includes income, education and life expectancy. Purchasing-power adjustments improve international comparison. No single index captures inequality, freedom, safety or environmental quality.

知识点 06/06 | 衡量生活水平最容易混淆的口径 | Common misconception

HDI 较高不表示每个分项都较高, 也不代表国内人人处于相同水平。

逐题精讲与评价拓展 | Exam workshop (共 3 项)

精讲 01/03 | 选择题: 区分最接近的概念 | Multiple-choice application

Question. Why is real GDP per head usually better than nominal GDP for comparing living standards over time?

正确答案 | Correct answer. It adjusts for both price changes and population, though it remains an average.

精讲 02/03 | 结构题: 把情境发展成完整分析 | Structured application

Question. Analyse two limitations of using GDP per head to compare countries.

参考答案 | Model answer. Exchange rates may not reflect local purchasing power, so equal dollar incomes buy different amounts. Informal and unpaid activity may be recorded differently. The average also hides

inequality, while pollution and leisure are omitted.

精讲 03/03 | 图表或计算题: 把方法落实到步骤 | Diagram or calculation application

Question. Country A has real GDP \$240bn and population 60m. Country B has real GDP \$180bn and population 30m. Calculate real GDP per head and compare material living standards. [4]

正确答案 | Correct answer. A: $\$240\text{bn} / 60\text{m} = \$4,000$ per head. B: $\$180\text{bn} / 30\text{m} = \$6,000$ per head. B has the higher average real output per person, although this alone does not prove every resident is better off because distribution and non-material factors are omitted.

第 26 课 | 2.3 贫困、人口与发展障碍

所属大章: 经济发展 | Economic development

知识点精讲 | Core knowledge

绝对贫困指基本需要难以满足; 相对贫困强调与社会正常生活水平的差距。人口快速增长可扩大劳动年龄人口和市场, 也可能压低人均资本、增加学校医疗负担与失业。低储蓄、基础设施不足、政治不稳定、初级产品依赖和债务均可形成发展陷阱。

深度展开与真题化应用 | In-depth development (共 8 个知识点)

知识点 01/08 | 贫困、人口与发展障碍: 贫困的两种口径 | Absolute and relative measures of poverty

Absolute poverty 指收入不足以满足 food、shelter 等基本需要; relative poverty 指收入显著低于社会通常水平。低收入 → 营养教育不足 → productivity 低 → 工资低, 形成 poverty cycle。税收转移、教育医疗和就业增长可能打破循环, 但政策有财政成本和激励问题。

知识点 02/08 | 贫困、人口与发展障碍: 人口数量之外还有结构 | Population size, structure and dependency

Birth rate、death rate、life expectancy 和 net migration 改变人口规模; age structure 与 dependency ratio 影响 labour supply、tax revenue 和公共支出。年轻人口可能带来未来劳动力优势, 也需要当前教育投入; 老龄化增加养老金和医疗压力, 但可能扩大养老服务需求。

知识点 03/08 | 贫困、人口与发展障碍中的主体、变量与判断对象 | Agents, variables and object of analysis

Absolute poverty 看 basic needs, relative poverty 看相对社会标准; population analysis 还要区分 birth rate、death rate、migration、age structure 与 dependency。

知识点 04/08 | 贫困、人口与发展障碍的因果机制 | How the mechanism works

low income → poor nutrition/education → low productivity → low wages → limited saving/investment → poverty persists。政策可在 education、health、credit 或 job creation 处打断循环。

知识点 05/08 | 贫困、人口与发展障碍的数字或现实情境 | Numerical or real-world application

工作年龄人口600万, 儿童250万, 老人150万, dependency ratio = $(250+150)/600 \times 100 = 66.7\%$ 。这不表示每100名工人实际供养66.7人, 因为工作年龄者并非全就业。

知识点 06/08 | 贫困、人口与发展障碍的关键英文表述 | Exam-ready explanation in English

Absolute poverty concerns inability to meet basic needs. Relative poverty compares a household with prevailing living standards. A high dependency ratio may increase pressure on workers and public services, but dependants can also contribute unpaid work and future labour.

知识点 07/08 | 贫困、人口与发展障碍最容易混淆的口径 | Common misconception

低 birth rate 不必然使人口立即下降; migration、death rate 和 population momentum 也会影响。

知识点 08/08 | 贫困、人口与发展障碍结论成立的条件 | When the conclusion may change

人口结果取决于年龄结构、人力资本、女性就业、工作机会和基础设施, 而非只看人数。

逐题精讲与评价拓展 | Exam workshop (共 3 项)

精讲 01/03 | 选择题: 区分最接近的概念 | Multiple-choice application

Question. Which change directly increases an ageing dependency ratio: higher birth rate, longer life expectancy with low fertility, net immigration of young workers, or later retirement?

正确答案 | Correct answer. Longer life expectancy with low fertility, because the elderly share rises relative to the working-age population.

精讲 02/03 | 结构题: 把情境发展成完整分析 | Structured application

Question. Discuss whether population growth benefits a developing economy.

参考答案 | Model answer. It can expand labour supply, demand and tax base, encouraging investment. A young population can become a demographic dividend. But education, housing and jobs must expand; otherwise unemployment, congestion and pressure on resources may rise.

精讲 03/03 | 图表或计算题: 把方法落实到步骤 | Diagram or calculation application

Question. A country has 18m people aged 0–14, 54m aged 15–64 and 12m aged 65+. Calculate the age dependency ratio. [3]

正确答案 | Correct answer. Dependent-age population = 18m + 12m = 30m. Dependency ratio = $30/54 \times 100 = 55.6\%$. This is an age-structure measure, not the ratio of non-workers to actual workers.

第 27 课 | 2.4 发展策略的评价

所属篇章: 经济发展 | Economic development

知识点精讲 | Core knowledge

教育、医疗、基础设施、微型信贷、贸易开放、FDI 与援助都可能有用。例: 修港口可降低出口运输成本, 但若政府腐败、维护资金不足, 项目效益会很低。评价要问: 谁受益? 资金来自哪里? 短期与长期效果? 环境是否可持续?

深度展开与真题化应用 | In-depth development (共 8 个知识点)

知识点 01/08 | 发展策略的评价: 政策通过生产率改变发展 | How policy raises productivity and development

Education raises skills and occupational mobility; healthcare reduces illness and improves effective labour supply; infrastructure lowers transport and communication costs; microfinance may support enterprise; property rights and stable institutions encourage investment。短期补贴能缓解贫困, 但长期发展通常要求生产能力与机会增加。

知识点 02/08 | 发展策略的评价: 评价预算、时滞与受益者 | Budgets, time lags and the distribution of benefits

同一笔预算投向高速公路会产生 opportunity cost, 例如少建学校。大型项目可能偏向城市, 农村贫困者未必直接受益。教育见效慢但外部收益广; 现金转移见效快却需持续财政来源。Foreign aid 和 FDI 可带来资金技术, 也可能有条件、利润外流和 dependency。

知识点 03/08 | 发展策略的评价中的主体、变量与判断对象 | Agents, variables and object of analysis

Education、healthcare、infrastructure、redistribution、aid 与 FDI 针对的 binding constraint 不同。比较政策要看 mechanism、time lag、cost、coverage 和 sustainability。

知识点 04/08 | 发展策略的评价的因果机制 | How the mechanism works

rural road investment → transport time/cost falls → farmers access larger markets → spoilage falls and revenue rises → saving/investment and rural living standards may improve。

知识点 05/08 | 发展策略的评价的数字或现实情境 | Numerical or real-world application

预算\$50m可建学校或发一年补贴。补贴立刻增加可支配收入; 学校需多年才转化为技能, 但收益可能持续更久。两者的 opportunity cost 与时间结构不同。

知识点 06/08 | 发展策略的评价的关键英文表述 | Exam-ready explanation in English

A development strategy should address the binding constraint. Education is weak if graduates have no jobs; roads are less useful without maintenance; foreign investment brings fewer benefits when local firms and workers cannot form linkages.

知识点 07/08 | 发展策略的评价最容易混淆的口径 | Common misconception

投入资金不等于 outcome 改善; 必须说明服务质量、覆盖人群与执行能力。

知识点 08/08 | 发展策略的评价结论成立的条件 | When the conclusion may change

“最佳”要求比较替代政策, 并结合该国最主要障碍作判断。

逐题精讲与评价拓展 | Exam workshop (共 2 项)

精讲 01/02 | 选择题: 区分最接近的概念 | Multiple-choice application

Question. Which policy most directly raises human capital: a tariff, teacher training, currency appreciation, or a maximum price?

正确答案 | Correct answer. Teacher training, because it can improve the quality of education and workers' future skills.

精讲 02/02 | 结构题: 把情境发展成完整分析 | Structured application

Question. Discuss whether foreign aid is the best way to promote development.

参考答案 | Model answer. Aid can finance health, education, disaster relief and infrastructure where domestic revenue is low. It may transfer expertise. However, tied aid, corruption, dependency and poor project selection reduce benefits. Trade access, governance or domestic taxation may be more sustainable.

第 28 课 | 2.5 指标题的比较框架

所属篇章：经济发展 | Economic development

知识点精讲 | Core knowledge

看到“哪个国家生活水平更高”，不要只比较一个 GDP 数字。依次问：是否是 real GDP？是否 per head？汇率是否反映购买力？收入是否平均分配？教育、寿命、饮水和环境怎样？一个资源出口国的人均 GDP 高，仍可能有很大贫富差距。

HDI 的读法：它把 income、education 与 life expectancy 组合，所以能补充 GDP；但它是国家平均值，无法显示一个城市很富而农村很穷。可直接写：HDI is broader than GDP per head, although both figures can hide inequality within a country.

深度展开与真题化应用 | In-depth development (共 6 个知识点)

知识点 01/06 | 指标题的比较框架：指标比较的四层结构 | A four-stage method for comparing indicators

第一层确认定义与单位；第二层描述 trend/difference；第三层用数据验证 relationship；第四层指出 exception 与 limitation。若比较 GDP，先确认 nominal or real、total or per head。若比较失业率，关注 labour-force 定义是否一致。

知识点 02/06 | 指标题的比较框架：平均值不代表每个人 | Why an average does not represent every household

GDP per head 上升可能与 median income 不同方向；HDI 改善也可能隐藏地区差异。数据只显示 correlation 时，不能直接写 A causes B。需要提出合理机制，并承认 reverse causation 或 third variable。

知识点 03/06 | 指标题的比较框架中的主体、变量与判断对象 | Agents, variables and object of analysis

Describe 只陈述数据，compare 必须涉及双方，calculate 展示公式，analyse 把数据证据与经济机制连接。题目动词决定答案任务。

知识点 04/06 | 指标题的比较框架的数字或现实情境 | Numerical or real-world application

失业率从4%到6%是上升2 percentage points，也是相对上升50%。题目若未指定，要用最清楚的表述，不能含混写“上升2%”。

知识点 05/06 | 指标题的比较框架的关键英文表述 | Exam-ready explanation in English

Use evidence selectively. A strong relationship answer states the overall pattern, cites two relevant comparisons, identifies an exception and offers an economic explanation. Correlation alone does not prove causation.

知识点 06/06 | 指标题的比较框架最容易混淆的口径 | Common misconception

不要只抄最高值与最低值；relationship 题需要跨多组数据，exception 也必须有数值证据。

逐题精讲与评价拓展 | Exam workshop (共 3 项)

精讲 01/03 | 选择题：区分最接近的概念 | Multiple-choice application

Question. Country X has twice the GDP of Y but four times the population. Which has higher GDP per head?

正确答案 | Correct answer. Country Y. X has only half Y's GDP per head when both totals are divided by population.

精讲 02/03 | 结构题: 把情境发展成完整分析 | Structured application

Question. Analyse a positive relationship between education spending and labour productivity shown in a table.

参考答案 | Model answer. Higher spending may improve teacher quality and access, raising workers' skills. Skilled workers use capital more effectively and produce more per hour. Cite supporting countries and an exception; productivity also depends on technology, health and spending efficiency.

精讲 03/03 | 图表或计算题: 把方法落实到步骤 | Diagram or calculation application

Question. Unemployment rises from 4% to 6%. State the change in percentage points and the percentage increase. [3]

正确答案 | Correct answer. The rate rises by 2 percentage points. Relative to the original rate, the percentage increase is $(6 - 4)/4 \times 100 = 50\%$. The two figures answer different questions.

Unit 6 | 国际贸易与全球化

International trade and globalisation

单元导读 | Unit introduction

本单元用 opportunity cost 重新解释 specialisation 与 comparative advantage, 再分析 free trade、protection、exchange rates、current account 与 globalisation。重点不是背诵“出口好、进口坏”，而是计算相对成本，追踪汇率改变相对价格后的数量反应，并比较消费者、企业、劳动者和政府的得失。

学习主线 | Learning route

比较优势题统一写成“生产 1 单位某商品放弃多少另一商品”；汇率题先翻译 appreciation/depreciation 对出口和进口价格的影响，再讨论 PED、合同和生产能力；保护主义评价要加入 retaliation 与 downstream costs。

English overview

Trade is based on comparative rather than absolute advantage. Exchange-rate changes first alter relative prices and only then affect quantities, depending on elasticity and time. Protection can support selected producers but may raise consumer prices, increase input costs and provoke retaliation.

本单元课时 | Lessons in this unit

第 29 课 | 2.1 专业化与比较优势；第 30 课 | 2.2 自由贸易与保护主义；第 31 课 | 2.3 汇率与经常账户；第 32 课 | 2.4 全球化；第 33 课 | 2.5 比较优势的数值法

第 29 课 | 2.1 专业化与比较优势

所属篇章：国际贸易与全球化 | International trade and globalisation

知识点精讲 | Core knowledge

Specialisation 让个人、企业或国家集中在机会成本较低的生产活动。关键是 comparative advantage（比较优势），不是绝对产量。即使 A 国两种产品都生产得更多，只要 A 放弃制造品生产食品的代价较低，它仍应专业化于食品，双方通过贸易可消费到 PPC 以外的组合。

深度展开与真题化应用 | In-depth development（共 7 个知识点）

知识点 01/07 | 专业化与比较优势：分工、专业化与交换 | Division of labour, specialisation and exchange

Specialisation 让工人、企业、地区或国家集中于较少任务，可提高 skill、速度和 machinery utilisation；风险是 boredom、dependence 和需求变化。国家贸易方向应由 comparative advantage 决定，即较低 opportunity cost，而非只看 absolute output。

知识点 02/07 | 专业化与比较优势：机会成本数值法 | Calculating comparative opportunity cost

A 每小时可生产 8 cloth 或 4 computers，因此 1 computer 成本为 2 cloth；B 可生产 6 cloth 或 6 computers，1 computer 成本为 1 cloth，所以 B 在 computers 有比较优势。反过来，A 的 1 cloth 成本为 0.5 computer，低于 B 的 1，因此 A 在 cloth 有比较优势。互利交换比率必须位于两国机会成本之间。

知识点 03/07 | 专业化与比较优势中的主体、变量与判断对象 | Agents, variables and object of analysis

Absolute advantage 比较相同资源下产量，comparative advantage 比较单位产品 opportunity cost。Specialisation 的收益来自 productivity、scale 与 exchange，代价来自 dependence 和 adjustment。

知识点 04/07 | 专业化与比较优势的因果机制 | How the mechanism works

specialisation → repeated practice and capital designed for a task → productivity rises → unit cost falls → output/trade may increase；同时 dependence 和 structural unemployment risk 可能上升。

知识点 05/07 | 专业化与比较优势的数值或现实情境 | Numerical or real-world application

A: 8 cloth 或 4 computers，1 computer=2 cloth；B: 6 cloth 或 6 computers，1 computer=1 cloth。B 在 computer 有比较优势，A 在 cloth 有比较优势。

知识点 06/07 | 专业化与比较优势的关键英文表述 | Exam-ready explanation in English

Comparative advantage explains potential gains from trade because resources are allocated to lower-opportunity-cost production. The model assumes transferable resources, acceptable transport costs and mutually beneficial terms of trade.

知识点 07/07 | 专业化与比较优势最容易混淆的口径 | Common misconception

不要用“谁产得多”判断 comparative advantage；必须比较 opportunity cost。

逐题精讲与评价拓展 | Exam workshop（共 2 项）

精讲 01/02 | 选择题：区分最接近的概念 | Multiple-choice application

Question. A country is less productive than another in both goods. Can it still have comparative advantage?

正确答案 | Correct answer. Yes. It can have a lower opportunity cost in one good even without an absolute advantage.

精讲 02/02 | 结构题: 把情境发展成完整分析 | Structured application

Question. Discuss whether greater specialisation benefits workers and consumers.

参考答案 | Model answer. Productivity and scale may lower prices and raise wages or choice. Workers gain expertise. However, repetitive work can reduce motivation, and demand changes may cause severe unemployment in specialised regions. Consumers become more dependent on supply chains.

第 30 课 | 2.2 自由贸易与保护主义

所属大章: 国际贸易与全球化 | International trade and globalisation

知识点精讲 | Core knowledge

自由贸易可降低价格、扩大选择、形成规模经济并增强竞争。保护主义包括 tariff、quota、subsidy 和 regulations；理由可为幼稚产业、就业、国家安全或反倾销，但代价通常是消费者价格更高、报复与低效率。高分答案必须同时写受保护企业和消费者，而非只写“有利/不利”。

深度展开与真题化应用 | In-depth development (共 8 个知识点)

知识点 01/08 | 自由贸易与保护主义: 自由贸易的收益与调整成本 | Benefits of free trade and adjustment costs

Free trade 扩大 market size, 促进 specialisation、competition、choice 和 economies of scale, 并降低进口价格。然而进口竞争会使缺乏竞争力的企业缩小, 造成 structural unemployment; 国家也可能依赖关键进口或承受外部冲击。

知识点 02/08 | 自由贸易与保护主义: 保护工具与利益相关者 | Protection instruments and stakeholder effects

Tariff 是进口税, 提高进口价格并产生政府收入; quota 直接限制数量, 可能产生配额租; subsidy 降低国内企业成本; embargo 禁止贸易。保护可支持 infant industry、employment、strategic goods 和 current account, 却提高消费者价格、削弱竞争, 并可能引发 retaliation。

知识点 03/08 | 自由贸易与保护主义中的主体、变量与判断对象 | Agents, variables and object of analysis

Tariff 改变进口价格并产生政府收入, quota 直接限制数量, subsidy 支持国内成本, embargo 禁止交易。每种工具对消费者、生产者与政府的影响不同。

知识点 04/08 | 自由贸易与保护主义的因果机制 | How the mechanism works

tariff on imported steel → import price rises → demand shifts towards domestic steel → local output/jobs may rise → car makers pay more → car prices rise and export competitiveness may fall → retaliation risk。

知识点 05/08 | 自由贸易与保护主义的数字或现实情境 | Numerical or real-world application

世界价\$100, tariff

20%, 进口价约\$120。若进口量从1m降至0.7m, 理论关税收入约\$14m; 但忽略行政成本和可能的逃税。

知识点 06/08 | 自由贸易与保护主义的关键英文表述 | Exam-ready explanation in English

Protection can support infant industries or strategic capacity, but it reduces competitive pressure and creates costs for users of imported inputs. Quotas restrict quantity directly; tariffs change price and raise government revenue.

知识点 07/08 | 自由贸易与保护主义最容易混淆的口径 | Common misconception

Tariff 不会机械地把 import quantity 降为零; 结果取决于需求、国内供给与税率。

知识点 08/08 | 自由贸易与保护主义结论成立的条件 | When the conclusion may change

结论要比较短期 adjustment benefits 与长期 efficiency costs, 并考虑 retaliation。

逐题精讲与评价拓展 | Exam workshop (共 2 项)

精讲 01/02 | 选择题: 区分最接近的概念 | Multiple-choice application

Question. Which group is most likely to lose directly from a tariff on imported food?

正确答案 | Correct answer. Domestic consumers, because they face higher prices and less choice, although local producers may gain.

精讲 02/02 | 结构题: 把情境发展成完整分析 | Structured application

Question. Discuss whether an infant industry should receive protection.

参考答案 | Model answer. Temporary protection may allow learning, scale economies and future competitiveness. It can preserve jobs and diversify the economy. But firms may become dependent and inefficient; consumers pay more and governments may be lobbied to extend protection. Conditions and a credible time limit matter.

第 31 课 | 2.3 汇率与经常账户

所属大章: 国际贸易与全球化 | International trade and globalisation

知识点精讲 | Core knowledge

汇率是一个货币用另一个货币表示的价格。货币 depreciation (贬值) 使出口在外国买家眼中更便宜、进口在国内更贵; 若需求有弹性, 净出口可能增加。但进口原料更贵也会推高成本型通胀, 而且短期合约会形成 J-curve。不要写“贬值一定改善贸易差额”。

深度展开与真题化应用 | In-depth development (共 8 个知识点)

知识点 01/08 | 汇率与经常账户: 汇率是两种货币的价格 | The exchange rate as the price of one currency

Appreciation 表示本币能兑换更多外币, 使 exports 在国外更贵、imports 对本国更便宜; depreciation 相反。浮动汇率受货币 demand 与 supply 影响: export demand、capital inflow 和高利率可能增加本币需求; import spending 与 capital outflow 增加本币供给。

知识点 02/08 | 汇率与经常账户: 数量反应决定最终结果 | Why quantity responses determine the final outcome

贬值首先改变价格, 但 current account 是否改善取决于 PED、替代品、质量、capacity 和 time lag。进口原材料昂贵会造成 cost-push inflation, 抵消出口价格优势。短期合同固定时数量反应慢, 可能出现 J-curve。

知识点 03/08 | 汇率与经常账户中的主体、变量与判断对象 | Agents, variables and object of analysis

Exchange rate 是两种货币的价格。Appreciation/depreciation 先改变 exports 和 imports 的相对价格, 随后才通过 PED、contracts 与 capacity 改变数量和 current account。

知识点 04/08 | 汇率与经常账户的因果机制 | How the mechanism works

currency depreciates → exports cheaper to foreigners/imports dearer domestically → X quantity may rise and M quantity fall → net exports and employment may rise; imported inputs become dearer → costs/inflation may rise。

知识点 05/08 | 汇率与经常账户的数字或现实情境 | Numerical or real-world application

£ 1=\$1.40变为 £ 1=\$1.20, 英镑 depreciation。售价 £ 100的商品由\$140降至\$120 (若英镑价不变); 英国买\$120商品由约 £ 85.71升至 £ 100。

知识点 06/08 | 汇率与经常账户的图表与数据读法 | Diagram and data interpretation

纵轴标本币价格或 exchange rate, 横轴标 quantity of currency。需求增加通常使本币升值; 随后分别分析出口价格、进口价格及数量反应, 不能从汇率直接跳到 current-account 结论。

知识点 07/08 | 汇率与经常账户的关键英文表述 | Exam-ready explanation in English

An exchange-rate change first alters relative prices. The current account changes only if buyers respond. Elasticity, contracts, production capacity and imported inputs determine the size and timing of the quantity response.

知识点 08/08 | 汇率与经常账户最容易混淆的口径 | Common misconception

Appreciation 不是“出口数量必然下降”; 先是相对价格改变, 再由 PED 决定数量。

逐题精讲与评价拓展 | Exam workshop (共 3 项)

精讲 01/03 | 选择题: 区分最接近的概念 | Multiple-choice application

Question. Who is most likely to gain from an appreciation: an exporter paid in foreign currency, a domestic tourist travelling abroad, or a producer competing with cheaper imports?

正确答案 | Correct answer. The tourist travelling abroad, because each unit of domestic currency buys more foreign currency.

精讲 02/03 | 结构题: 把情境发展成完整分析 | Structured application

Question. Discuss whether depreciation will reduce a current-account deficit.

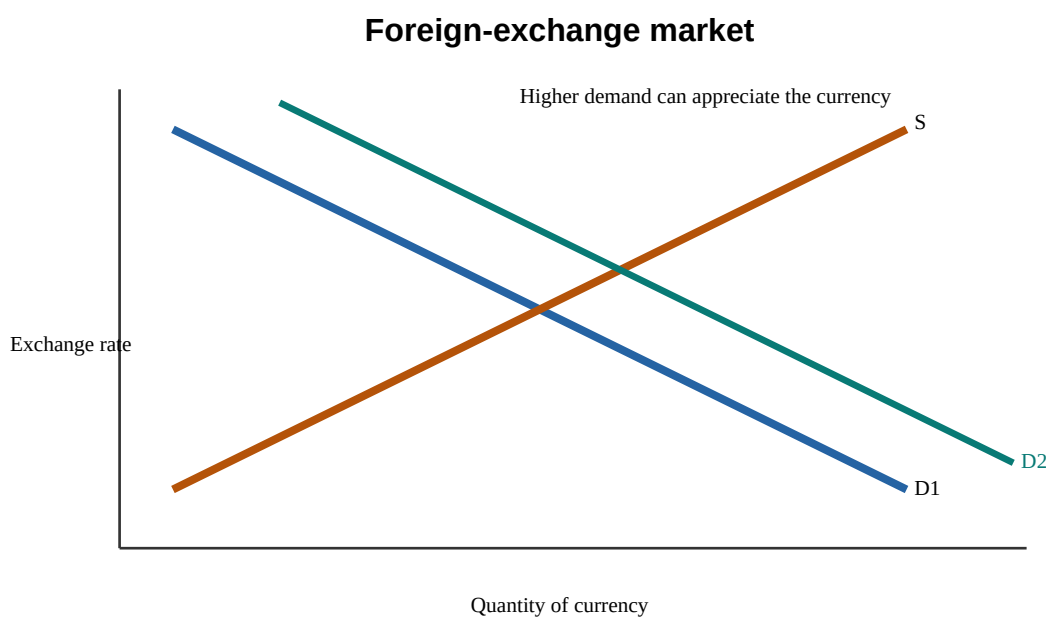
参考答案 | Model answer. Cheaper exports and dearer imports may raise export receipts and reduce import spending. Yet demand may be inelastic, firms may lack capacity, and imported raw materials raise costs. Short-run contracts can delay adjustment. Success depends on elasticities and time.

精讲 03/03 | 图表或计算题: 把方法落实到步骤 | Diagram or calculation application

Question. The exchange rate changes from £1=\$1.40 to £1=\$1.20. Identify the change and calculate the new dollar price of a £250 export. [3]

正确答案 | Correct answer. The pound depreciates because each pound buys fewer dollars. The export price changes from \$350 to \$300. Whether export revenue rises depends on the response of foreign quantity demanded.

图表推演 | Diagram analysis



考试图表必须标出坐标轴、曲线名称、新旧均衡和变化方向。文字解释需说明原因、曲线移动以及价格或数量结果。

第 32 课 | 2.4 全球化

所属大章: 国际贸易与全球化 | International trade and globalisation

知识点精讲 | Core knowledge

全球化提高资本、信息、商品和人员流动, 可能带来 FDI、就业和技术转移, 也可能令弱劳动者面临跨国竞争、利润外流和文化/环境压力。判断题应结合一个国家的技能、制度与产业结构。

深度展开与真题化应用 | In-depth development (共 7 个知识点)

知识点 01/07 | 全球化: 全球化不只是贸易增加 | Globalisation beyond an increase in trade

Globalisation 指商品、服务、资本、技术、信息和生产跨国联系增强。Multinational companies 通过 FDI 建厂, 可能带来 capital、employment、training、tax revenue 和 access to export markets; 东道国也可能面对 profit repatriation、tax avoidance、environmental damage 和对本地企业的竞争。

知识点 02/07 | 全球化: 不同群体获得不同结果 | Why globalisation creates different stakeholder outcomes

消费者可能因进口和竞争获得更低价格与更多选择; 出口企业进入更大市场; 部分工人获得岗位, 另一些工人因 offshoring 失业。来源国企业可降低成本, 却可能减少本国某些就业。评价必须分别讨论 consumers、workers、firms、government 和 environment。

知识点 03/07 | 全球化中的主体、变量与判断对象 | Agents, variables and object of analysis

Globalisation 包括 trade、FDI、capital、technology、information 与 labour 的跨国联系。MNC 是参与者之一, 不能把两者当作同义词。

知识点 04/07 | 全球化的因果机制 | How the mechanism works

MNC builds factory → capital and jobs enter host economy → workers receive income/training → local suppliers gain orders → tax revenue may fund services; 若 inputs imported and profits repatriated, local multiplier 较小。

知识点 05/07 | 全球化的数字或现实情境 | Numerical or real-world application

工厂销售\$100m, 但进口投入\$45m、利润汇回\$25m, 本地工资税收采购只有\$30m。Gross sales 不能代表留在东道国的全部收益。

知识点 06/07 | 全球化的关键英文表述 | Exam-ready explanation in English

Technology transfer and access to global markets can raise productivity. However, bargaining power, tax rules, environmental regulation and local linkages determine how gains are shared. Globalisation creates winners and losers within countries.

知识点 07/07 | 全球化结论成立的条件 | When the conclusion may change

短期 job creation 与长期 capability building 要分开, 尤其关注技能转移和本地供应链。

逐题精讲与评价拓展 | Exam workshop (共 2 项)

精讲 01/02 | 选择题: 区分最接近的概念 | Multiple-choice application

Question. Which is a likely host-country benefit of FDI: guaranteed equal income, technology transfer, no competition, or elimination of imports?

正确答案 | Correct answer. Technology transfer, although its extent depends on training and links with domestic firms.

精讲 02/02 | 结构题: 把情境发展成完整分析 | Structured application

Question. Discuss whether MNCs improve living standards in developing economies.

参考答案 | Model answer. They may create employment, wages, training, exports and tax revenue.

Competition and technology can raise productivity. Conversely, profits may leave, local firms may close, jobs may be insecure and pollution may rise. Regulation and job quality determine the net effect.

第 33 课 | 2.5 比较优势的数值法

所属大章: 国际贸易与全球化 | International trade and globalisation

知识点精讲 | Core knowledge

若 A 国一小时产 8 件衣服或 4 台电脑, 生产 1 台电脑的机会成本是 2 件衣服; B 国一小时产 6 件衣服或 6 台电脑, 生产 1 台电脑的机会成本是 1 件衣服。因此 B 在电脑有比较优势, A 在衣服有比较优势。比较的是比率, 不能只看“谁生产得更多”。

汇率题先说明报价方式。若 1 USD 从 7.0 CNY 升至 7.3

CNY, 人民币对美元贬值; 中国出口对美国买家更便宜, 但进口原材料更贵。紧接着加一句条件: The effect on the current account depends on price elasticities and the time needed for contracts to change.

深度展开与真题化应用 | In-depth development (共 6 个知识点)

知识点 01/06 | 比较优势的数值法: 先统一“每一单位”的方向 | Using a consistent per-unit opportunity cost

若表格给的是相同资源下最大产量, 用“另一商品产量 ÷ 本商品产量”求一单位机会成本。A 可产 12 wheat 或 6 cars, 则 1 car=2 wheat, 1 wheat=0.5 car。每个国家都算两行, 再横向比较同一商品; 机会成本较低者有 comparative advantage。

知识点 02/06 | 比较优势的数值法: 检验交换比率是否互利 | Testing whether the terms of trade benefit both countries

Terms of trade 必须介于两国国内机会成本之间。若 A 生产 1 car 放弃 2 wheat, B 放弃 4 wheat, 国际交换率为 1 car:3 wheat, 则 A 出口 car 可换 3 wheat, 超过自行转产得到的 2; B 进口 car 只付 3 wheat, 少于自行生产放弃的 4。

知识点 03/06 | 比较优势的数值法中的主体、变量与判断对象 | Agents, variables and object of analysis

对每个国家分别计算 1 unit X 的 Y opportunity cost 和 1 unit Y 的 X opportunity cost, 再横向比较同一商品。两个单位成本互为倒数。

知识点 04/06 | 比较优势的数值法的数字或现实情境 | Numerical or real-world application

A:12 wheat或6 cars, 所以1 car=2 wheat; B:8 wheat或2 cars, 所以1 car=4 wheat。A出口car。若1 car换3 wheat, A得到超过2, B支付少于4, 双方获益。

知识点 05/06 | 比较优势的数值法最容易混淆的口径 | Common misconception

分子分母颠倒会同时错判两个产品。答案必须写单位, 例如 wheat per car。

知识点 06/06 | 比较优势的数值法的综合应用 | Integrated application

A 的 1 car=2 wheat, B 的 1 car=4 wheat; 若 terms of trade 为 1 car:3 wheat, A 出口 car 得到 3 wheat, B 进口 car 只付 3 wheat, 双方都优于国内机会成本。

逐题精讲与评价拓展 | Exam workshop (共 3 项)

精讲 01/03 | 选择题: 区分最接近的概念 | Multiple-choice application

Question. Country A gives up 2 rice for one phone; B gives up 5. Which exchange rate benefits both?

正确答案 | Correct answer. One phone for any amount strictly between 2 and 5 rice, such as 3 rice.

精讲 02/03 | 结构题: 把情境发展成完整分析 | Structured application

Question. Using opportunity-cost data, explain why two countries may gain from trade.

参考答案 | Model answer. The country with the lower opportunity cost should specialise relatively more in that product. Combined output can rise because resources are used where sacrifice is smaller. At terms of trade between the two costs, the exporter receives more than domestically and the importer gives up less.

精讲 03/03 | 图表或计算题: 把方法落实到步骤 | Diagram or calculation application

Question. Country A can produce 12 wheat or 6 cars; B can produce 8 wheat or 2 cars. Calculate the opportunity cost of one car in each country and identify comparative advantage. [5]

正确答案 | Correct answer. A: one car costs $12/6 = 2$ wheat. B: one car costs $8/2 = 4$ wheat. A has comparative advantage in cars because its opportunity cost is lower; B has comparative advantage in wheat because one wheat costs $2/8 = 0.25$ car, compared with $6/12 = 0.5$ car in A.